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OPTIMALITY CONDITIONS AND NUMERICAL ALGORITHMS FOR A CLASS OF LINEARLY CONSTRAINED MINIMAX OPTIMIZATION PROBLEMS*

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Abstract. It is well known that there have been many numerical algorithms for solving nonsmooth minimax problems; however, numerical algorithms for nonsmooth minimax problems with joint linear constraints are very rare. This paper aims to discuss optimality conditions and develop practical numerical algorithms for minimax problems with joint linear constraints. First, we use the properties of proximal mapping and the KKT system to establish optimality conditions. Second, we propose a framework of an alternating coordinate algorithm for the minimax problem and analyze its convergence properties. Third, we develop a proximal gradient multistep ascent descent method (PGmsAD) as a numerical algorithm and demonstrate that the method can find an ϵ-stationary point for this kind of nonsmooth problem in $\mathcal{O}\left(\epsilon^{-2}\log\epsilon^{-1}\right)$ iterations. Finally, we apply PGmsAD to generalized absolute value equations, generalized linear projection equations, and linear regression problems, and we report the efficiency of PGmsAD on large-scale optimization.

Key words. minimax optimization, proximal mapping, proximal gradient multi-step ascent descent method, iteration complexity, generalized absolute value equations, linear regression, generalized linear projection equations

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1. Introduction.

Linear equation constrained optimization problems arise in numerous scientific and engineering applications including signal processing [39] and distributed optimization [41]. Many studies, such as [30, 56], have developed the optimality theory and numerical algorithms for the linear equation constraint minimization problem, which is one of the most important constraint problems. The main purpose of this paper is to build optimality conditions and numerical algorithms for the minimax optimization problem with joint linear constraints,

$$\begin{array}{ll} \min\limits_{x\in\Re^n}\max\limits_{y\in\Re^m} & f(x,y)=\varphi(x)+g(x)+x^TKy-h(y)-\psi(y) \\ \text{subject to} & Ax+By+c=0, \end{array} \tag{1.1}$$

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where $g : \Re^n \to \Re$ and $h : \Re^m \to \Re$ are real-valued smooth functions, $\varphi : \Re^n \to \overline{\Re}$ and $\psi : \Re^m \to \overline{\Re}$ are extended real-valued proper lower semicontinuous convex functions (here $\overline{\Re} = \Re \cup \{+\infty\}$), $K \in \Re^{n\times m}$, $A \in \Re^{q\times n}$, $B \in \Re^{q\times m}$, and $c \in \Re^q$.

1.1. Applications. Minimax problems (1.1) play a significant role in optimization since many interesting linear equations can be converted to equivalent linearly constrained minimax problems. Traditional methods for solving linear equations include the least squares method [7], the fixed point method [1], and the Newton method [2]. However, these classical numerical algorithms cannot handle equations with absolute values or projection operators, such as generalized absolute value equations (GAVEs) [66] and generalized linear projection equations (GLPEs) [25]. Recently, numerical algorithms have been proposed only to solve such equations with special structures, such as generalized Newton methods [46], neural networks [62], etc. It is noteworthy that these intractable linear system problems can be translated into equivalent linearly constrained minimax problems.

Generalized absolute value equations (GAVEs). The GAVE is a popular nonsmooth NP-hard problem in the form of

$$Ax + B|x| = b, \tag{1.2}$$

where $A,\ B \in \Re^{m\times n},\ b \in \Re^m$, and for $x = (x_1, \ldots, x_n) \in \Re^n$, $|x| := (|x_1|, \ldots, |x_n|) \in \Re^n_+$. As an important tool in the field of optimization, GAVE is widely used to solve problems in diverse fields, including nonnegative constrained least squares problems, quadratic programming, complementarity problems, and bimatrix games (e.g., [20, 24, 48, 65]). In this paper, we prove that GAVE is equivalent to a smooth nonseparable linearly constraint convex-concave minimax problem as follows:

$$\begin{array}{ll} \min\limits_{x\in\Re^n_+} \max\limits_{z\in\Re^m_+, y\in\Re^m} & (b-(A+B)x)^T y \\ \text{subject to} & x-(B-A)^T y - z = 0. \end{array}$$

Moreover, we also focus on the well-known linear regression problems with joint linear constraints and strongly convex strongly concave quadratic objective functions.

Generalized linear projection equations (GLPEs). As one of the important techniques for solving constrained optimization problems, GLPE has attracted extensive attention in linear variational inequalities, fixed point problems, bimatrix equilibrium points, and traffic network modeling (e.g., [20, 26, 63]). We are concerned with solving GLPEs of the following form:

$$Ax + Bx_K = b, \text{ subject to } x_K = P_K(x), \tag{1.3}$$

where $A,\ B \in \Re^{m\times n},\ b \in \Re^m$, $K \subset \Re^n$ is a closed convex set and for any $x \in \Re^n$, $P_K(x)$ represents the projection of x onto K. If K is a cone, GLPE can be translated into a nonsmooth nonseparable linearly constrained convex-concave minimax problem,

$$\begin{array}{ll} \min\limits_{x\in\Re^n} \max\limits_{y\in\Re^m, z\in\Re^n} & \delta_K(x) + (b-(A+B)x)^T y - \delta_{K^\circ}(z) \\ \text{subject to} & x - A^T y - z = 0, \end{array}$$

where K° represents the polar cone of K and $\delta_K(x)$ is an indicator function.

Both GAVE and GLPE are two important specific examples of the general minimax optimization problem (1.1). The numerical algorithms proposed for (1.1) can also be efficiently applied for GAVE and GLPE, as shown in section 5.

1.2. Related works. There are many effective algorithms that can be solved when $A = 0$ or $B = 0$ in linearly constrained minimax problems (1.1), such as the alternating coordinate method [4], the first-order primal-dual algorithm [13], and the multistep gradient descent ascent method [55]. However, when $A \neq 0$ and $B \neq 0$, since the variables x, y in the constraint are coupled, (1.1) cannot be solved by the above conventional algorithms. It is noted that, by introducing the Lagrange multiplier and establishing the Lagrange function, we can solve nonlinear programming problems effectively and quickly (see [8]). Inspired by this, we hope to design an effective algorithm to solve the joint linearly constrained minimax problem by establishing the Lagrange function of (1.1). Since φ and ψ in (1.1) are nonsmooth, the proximal mapping technique is used to deal with nonsmooth terms.

The study of algorithms for solving minimax problems is quite active. For the case when variables x and y have separable constraints, there are many publications about constructing and analyzing numerical algorithms for the minimax problem, such as [4, 13, 15, 14, 16, 40, 51, 53, 55, 43, 64]. Just recently, proximal projection gradient-type algorithms have been studied deeply for nonsmooth minimax optimization problems. Valkonen [59] proposed an extension of the modified primal-dual hybrid gradient method, due to Chambolle and Pock [13]. Mokhtari et al. [50] proposed algorithms admitting a unified analysis as approximations of the classical proximal point method for solving saddle point problems. Dai et al. [21] proposed a semi-proximal alternating coordinate method for solving such a structured convex-concave minimax problem and proved the global convergence as well as the linear rate of convergence. Further, a primal-dual proximal splitting (PDPS) method proposed by Clason et al. [19] and an inexact primal-dual smoothing (IPDS) framework studied by Hien et al. in [38] are used to solve minimax problems with nonsmooth terms, and the convergence is established provided with some Lipschitz properties. He [31] proposed a hybrid proximal extragradient framework where subproblems require solving the subdifferential of nonsmooth functions. A Nesterov accelerated variant is used to approximately solve the prox subproblems and finds an (ϵ, ϵ)-saddle point in $\mathcal{O}\left(\epsilon^{-2}\right)$ iterations. Recently, Hamedani [29] proposed a primal-dual algorithm for the nonsmooth convex-concave minimax problem and achieved an ergodic convergence rate of function value with $O(1/k)$. However, these methods are not valid when x and y interact in constraints.

Recently, effective numerical algorithms have been proposed for the smooth problem (1.1) with $\varphi \equiv 0$ and $\psi \equiv 0$. For minimax problems with joint linear inequality constraints, Tsaknakis et al. [34] proposed a multiplier gradient descent method and studied its convergence properties. However, they did not analyze iteration complexities of the method in [34]. Dai and Zhang established the optimal conditions for general constrained minimax optimization problems in [22] and proposed an augmented Lagrangian (AL) method for minimax problems with equality constraints in [23], where the AL method has superlinear convergence rate if the penalty parameter is increasing to infinity. Goktas and Greenwald conducted a series of studies on minimax problems with joint nonlinear inequality constraints. They developed the max-oracle gradient descent (MOGD) method [27] and the Lagrangian gradient descent ascent (LGDA) method [28] and proved that these methods can find an ϵ-Stackelberg equilibrium in $\mathcal{O}\left(\epsilon^{-2}\right)$ iterations. However, for nonsmooth linearly nonseparable constraint minimax problems of the form (1.1), to the best of our knowledge, numerical algorithms are quite rare. Therefore, we hope to develop numerical algorithms for nonsmooth linearly nonseparable constraint minimax problems, establish their iteration complexities, and apply them to solve linear equations with absolute values or projection operators.

1.3. Contribution. In this paper, we focus on optimality conditions and numerical algorithms for nonsmooth linearly nonseparable constraint convex-concave minimax problems (1.1). Our contribution mainly consists of the following three aspects. First, we present an alternating coordinate ascent descent algorithm framework and prove the convergence of iterations. Our key idea is to use the Lagrange function of (1.1) to deal with the nonseparable constraint in (1.1). Under mild conditions (Assumption 2.1, i.e., some Lipschitz properties), (1.1) can be reformulated as the following unconstrained nonsmooth minimax problem:

$$\min_{x,\lambda}\max_{y}\varphi(x)+g(x)+x^TKy-h(y)-\psi(y)+\langle\lambda,Ax+By+c\rangle,$$

where $\lambda\in\Re^q$ is a Lagrange multiplier. For the above unconstrained nonsmooth problem, inspired by the success of proximal gradient methods in nonsmooth optimization, we use the proximal mapping technique to deal with nonsmooth terms and define the stationary point of (1.1) and establish equivalence conditions and properties of the stationary point. The convergence of the algorithm framework depends on finding an ϵ-stationary point of the subproblem with respect to y.

Second, in order to efficiently solve subproblems in the alternating coordinate ascent descent algorithm, we develop a proximal gradient multi-step ascent descent method and show that this algorithm can find an ϵ-stationary point in $\mathcal{O}\left(\epsilon^{-2}\log\epsilon^{-1}\right)$ iterations. The basic idea of this algorithm is to incorporate a multistep proximal acceleration scheme into the alternating coordinate ascent descent algorithm framework. We demonstrate that the algorithm ensures that the subproblem with respect to y achieves an ϵ-stationary point under the convexity and the Lipschitz property of the objective function. Further, we can establish the same iterative complexity of the algorithm as in the unconstrained minimax problem [55, 43].

Finally, we focus on solving several classes of linear equations, including generalized absolute value equations, linear regression, and generalized linear projection equations. We prove that GAVE and GLPE are equivalent to nonseparable linearly constrained convex- concave minimax problems and use the proposed numerical algorithm to solve these popular linear equations. To the best of our knowledge, this is the first time that such a numerical algorithm has been developed for GAVE and GLPE without any special structure in the literature.

1.4. Organization. The paper is organized as follows. In section 2, we define the stationary point of nonsmooth linearly constrained convex-concave minimax problems and establish optimality conditions. In section 3, we develop an alternating coordinate ascent descent method, and any accumulation point of the sequences generated by this method is a stationary point of (1.1). In section 4, we construct a proximal gradient multistep ascent descent method for (1.1) and prove that this algorithm generates a sequence converging to an ϵ-stationary point within $\mathcal{O}(\epsilon^2\log\epsilon^{-1})$ iterations. In section 5, we apply the proximal gradient multistep ascent descent method to generalized absolute value equations and linear regression problems and show the efficiency of our algorithm. Some remarks are made in the last section.

1.5. Notation. The following notation is used throughout this paper. $\Re^n_+$ denotes an n-dimensional positive octant cone. We use $\|\cdot\|$ to represent the Euclidean norm and its induced matrix norm and $\mathbf{B}\,(x,d)$ to denote a closed ball of radius $d>0$ centered at x. For any $x\in\Re^n$, $\nabla h(x)$ and $\partial g(x)$ represent the gradient of the smooth function h at x and the subderivative of the nonsmooth function g at x, respectively. The conjugate function of $f:\Re^n\to\Re^m$ is defined as $f^*:\Re^m\to\Re^n$. $\operatorname{dom} h$ denotes the domain of the proper lower semicontinuous function h. $\mathrm{dist}(x, C)$ represents the distance from the point $x \in \Re^n$ to the set $C \subseteq \Re^n$.

It is said that $h : \Re^n \to \Re$ is an L_h-smooth function, if h is continuously differentiable and its gradient mapping is Lipschitz continuous with Lipschitz constant $L_h > 0$:

$$\|\nabla h(z') - \nabla h(z)\| \le L_h \|z' - z\|, \quad \forall z', z \in \Re^n.$$

For $\lambda > 0$, the proximal mapping for σ, denoted as $\mathrm{prox}_{\lambda\sigma}$, is defined by

$$\mathrm{prox}_{\lambda\sigma}(z) = \operatorname*{argmin}_{z' \in \mathrm{dom}\,\sigma} \left\{ \lambda\sigma(z') + \frac{1}{2}\|z' - z\|^2 \right\}.$$

Define, for $L > 0$,

$$T_L^{h,\sigma}(z) = \mathrm{prox}_{L^{-1}\sigma}(z - L^{-1}\nabla h(z)) \text{ and } G_L^{h,\sigma}(z) = L(z - T_L^{h,\sigma}(z)).$$

2. Properties of the nonsmooth minimax optimization problem. In this section, we analyze properties of the nonsmooth minimax optimization problem (1.1). Define the feasible region of problem (1.1) by

$$C = \{(x, y) \in \Re^n \times \Re^m : Ax + By + c = 0\}. \tag{2.1}$$

First, we propose some basic assumptions about functions g, h, φ, and ψ, which are common in nonsmooth minimax optimization problems, such as those in [11, 17, 33, 67]. Denote $\varphi_g := \varphi + g$ and $\psi_h := h + \psi$.

Assumption 2.1. Let functions $g : \Re^n \to \Re, h : \Re^m \to \Re$, $\varphi : \Re^n \to \overline{\Re}$ and $\psi : \Re^m \to \overline{\Re}$ satisfy the following conditions.

A1 Functions g and h are continuously differentiable convex with Lipschitz continuous gradients; i.e., there exist constants $L_g > 0$ and $L_h > 0$ such that for any $(x', y'), (x, y) \in \Re^n \times \Re^m$,

$$\|\nabla g(x') - \nabla g(x)\| \le L_g\|x' - x\|, \|\nabla h(y') - \nabla h(y)\| \le L_h\|y' - y\|.$$

A2 Functions φ and ψ are proper lower semicontinuous convex functions.

Define the Lagrange function of (1.1),

$$\mathcal{L}(x, y, \lambda) = \varphi(x) + g(x) + x^T K y - h(y) - \psi(y) + \langle \lambda, Ax + By + c\rangle,$$

where $\lambda \in \Re^q$ is a Lagrange multiplier. It follows from Assumption 2.1(**A1)**, for any given $y \in \Re^m$ and $y' \in \Re^m$, that

$$-h(y') \ge -h(y) + \nabla_y h(y)(y' - y) - \frac{L_h}{2}\|y' - y\|^2.$$

For $\mu \ge L_h$, define

$$\begin{aligned} Q(x, z, y, \lambda) &= x^T K y - h(y) - \psi(y) + \langle \lambda, Ax + By + c\rangle - \frac{\mu}{2}\|y - z\|^2, \\ \theta_0(x, z, \lambda) &= \max_y Q(x, z, y, \lambda), \end{aligned} \tag{2.2}$$

where the function $y \to Q(x, z, y, \lambda)$ is μ-strongly concave, which implies that the problem $\max_y Q(x, z, y, \lambda)$ has a unique solution, denoted by

$$y_*(x, z, \lambda) = \mathrm{argmax}\,\{Q(x, z, y, \lambda) : y \in \Re^m\}.$$

For a fixed point (x, z, λ), we define functions with respect to y as follows:

$$\begin{aligned} \psi_h(y) &= h(y) + \psi(y),\ \varphi_g(x) = \varphi(x) + g(x), \\ \theta(x,z,\lambda) &= g(x) + \theta_0(x,z,\lambda),\ \theta_\varphi(x,z,\lambda) = \varphi_g(x) + \theta_0(x,z,\lambda). \end{aligned} \tag{2.3}$$

If Assumption 2.1 is satisfied, then, we have

$$0 \in -[K^T x + B^T \lambda - \mu(y_*(x,z,\lambda) - z)] + \partial\psi_h(y_*(x,z,\lambda)). \tag{2.4}$$

To ensure sufficient gradient descent in y, inspired by the analysis of the minimization problem [3], in the following proposition, we will prove that y_* is Lipschitz continuous.

Proposition 2.1. *Let Assumption* 2.1 *be satisfied and* $\mu \ge L_h$. *Then for any* (x^0, z^0, λ^0), $(x^1, z^1, \lambda^1) \in \operatorname{dom}\theta_0$, *there exist* $y_*(x^0, z^0, \lambda^0) \in \operatorname{argmax}_y Q(x^0, z^0, y, \lambda^0)$ *and* $y_*(x^1, z^1, \lambda^1) \in \operatorname{argmax}_y\ Q(x^1, z^1, y, \lambda^1)$ *such that*

$$\|y_*(x^0,z^0,\lambda^0) - y_*(x^1,z^1,\lambda^1)\| \le \mu^{-1}[\|K\|\|x^0 - x^1\| + \mu\|z^0 - z^1\| + \|B\|\|\lambda^0 - \lambda^1\|]. \tag{2.5}$$

The function $\theta(x,z,\lambda)$ *is continuously differentiable at any* $(x,z,\lambda) \in \operatorname{dom}\theta$ *with*

$$\nabla\theta(x,z,\lambda) = \begin{pmatrix} \nabla g(x) + A^T\lambda + K y_*(x,z,\lambda) \\ \mu(-z + y_*(x,z,\lambda)) \\ Ax + By_*(x,z,\lambda) + c \end{pmatrix}, \tag{2.6}$$

and

$$\|\nabla\theta(x^0,z^0,\lambda^0) - \nabla\theta(x^1,z^1,\lambda^1)\| \le \frac{\gamma}{\mu}\|(x^0,z^0,\lambda^0) - (x^1,z^1,\lambda^1)\| \tag{2.7}$$

for any $(x^0,z^0,\lambda^0) \in \operatorname{dom}\theta$ *and* $(x^1,z^1,\lambda^1) \in \operatorname{dom}\theta$, *where*

$$\begin{aligned} \gamma = \sqrt{3}\max\Big\{ &\sqrt{(L_g\mu + \|K\|^2)^2 + \gamma_0 + \mu^2\|K\|^2}, \sqrt{4\mu^2 + \|K\|^2 + \|B\|^2}, \\ &\sqrt{\gamma_0 + \|B\|^4 + \mu^2\|B\|^2}\Big\}, \end{aligned}$$

and $\gamma_0 = (\mu\|A\| + \|K\|\|B\|)^2$.

Proof. From the definition of $y_*(x,z,\lambda)$ and the μ-strong concavity of the function $y \to -Q(x,z,y,\lambda)$, there exist $y^0 = y_*(x^0,z^0,\lambda^0) \in \operatorname{argmax}_y Q(x^0,z^0,y,\lambda^0)$ and $y^1 = y_*(x^1,z^1,\lambda^1) \in \operatorname{argmax}_y Q(x^1,z^1,y,\lambda^1)$ such that for any $y \in \Re^m$,

$$-Q(x^0,z^0,y,\lambda^0) \ge -Q(x^0,z^0,y^0,\lambda^0) + \frac{\mu}{2}\|y - y^0\|^2 \tag{2.8}$$

and

$$-Q(x^1,z^1,y,\lambda^1) \ge -Q(x^1,z^1,y^1,\lambda^1) + \frac{\mu}{2}\|y - y^1\|^2. \tag{2.9}$$

Replacing $y = y^1$ in (2.8) yields

$$-Q(x^0,z^0,y^1,\lambda^0) \ge -Q(x^0,z^0,y^0,\lambda^0) + \frac{\mu}{2}\|y^1 - y^0\|^2. \tag{2.10}$$

Replacing $y = y^0$ in (2.9) yields

$$-Q(x^1, z^1, y^0, \lambda^1) \geq -Q(x^1, z^1, y^1, \lambda^1) + \frac{\mu}{2}\|y^0 - y^1\|^2. \tag{2.11}$$

Adding both sides of (2.10) and (2.11), we obtain

$$\begin{aligned}&\langle K^T x^0, y^0\rangle - \langle K^T x^0, y^1\rangle + \langle K^T x^1, y^1\rangle - \langle K^T x^1, y^0\rangle + \langle \lambda^1 - \lambda^0, B(y^1 - y^0)\rangle \\ &\quad + \frac{\mu}{2}\left[\|y^1 - z^0\|^2 - \|y^0 - z^0\|^2 + \|y^0 - z^1\|^2 - \|y^1 - z^1\|^2\right] \geq \mu\|y^0 - y^1\|^2,\end{aligned}$$

or equivalently

$$\langle K(y^1 - y^0), x^1 - x^0\rangle + \langle \lambda^1 - \lambda^0, B(y^1 - y^0)\rangle + \mu(z^0 - z^1)^T(y^0 - y^1) \geq \mu\|y^0 - y^1\|^2. \tag{2.12}$$

Therefore, we have from (2.12) that

$$\mu\|y^0 - y^1\|^2 \leq \|K\|\|x^0 - x^1\|\|y^0 - y^1\| + \|B\|\|\lambda^0 - \lambda^1\|\|y^0 - y^1\| + \mu\|z^0 - z^1\|\|y^0 - y^1\|,$$

which implies the inequality (2.5).

Now we prove that θ is differentiable at $(x, z, \lambda) \in \operatorname{dom}\theta$. Since $y_*(x, z, \lambda)$ is Lipschitz continuous, for fixed positive number ε_0, for any $(x, z, \lambda) = \operatorname{dom}\theta$, from the lower semicontinuity of ψ, there exists a positive number $\delta > 0$, for $(x', z', \lambda') \in \mathbf{B}_\delta(x, z, \lambda)$,

$$\psi(y_*(x', z', \lambda')) \geq \psi(y_*(x, z, \lambda)) - \varepsilon_0.$$

Let D be a nonempty convex compact set satisfying $y_*(\mathbf{B}_\delta(x, z, \lambda)) \subseteq D$. Then for $(x', z', \lambda') \in \mathbf{B}_\delta(x, z, \lambda)$, one has the following expression:

$$\begin{aligned}\theta(x', z', \lambda') = \max_{y,s}\ & g(x') + (x')^T K y - h(y) - \psi(y) + \langle \lambda', Ax' + By + c\rangle \\ & -\frac{\mu}{2}\|y - z'\|^2 - s, \\ (y, s) \in & \left\{D \times [-\varepsilon_0, \varepsilon_0]\right\} \cap \operatorname{epi}\psi.\end{aligned}$$

Noting that the set $\{D \times [-\varepsilon_0, \varepsilon_0]\} \cap \operatorname{epi}\psi$ is a nonempty convex compact set, we may employ the famous Danskin theorem and obtain that θ is directionally differentiable at (x, z, λ) with

$$\begin{aligned}\theta'(x, z, \lambda; d_x, d_z, d_\lambda) = {} & (\nabla g(x) + A^T\lambda + K y_*(x, z, \lambda))^T d_x - \mu(z - y_*(x, z, \lambda))^T d_z \\ & + (Ax + By_*(x, z, \lambda) + c)^T d_\lambda.\end{aligned}$$

This implies that θ is differentiable at (x, z, λ) and the formula (2.6) holds. From this formula and (2.5), we have

$$\begin{aligned}&\left\|\nabla\theta(x^0, z^0, \lambda^0) - \nabla\theta(x^1, z^1, \lambda^1)\right\|^2 \\ &\quad = \left\|\nabla g(x^0) - \nabla g(x^1) + A^T(\lambda^0 - \lambda^1) + K[y_*(x^0, z^0, \lambda^0) - y_*(x^1, z^1, \lambda^1)]\right\|^2 \\ &\qquad + \mu^2\left\|-(z^0 - y^0) + (z^1 - y^1)\right\|^2 + \left\|A(x^0 - x^1) + B[y_*(x^0, \lambda^0) - y_*(x^1, \lambda^1)]\right\|^2\end{aligned}$$

$$
\begin{aligned}
&\leq [L_g\|x^1 - x^0\| + \|A\|\|\lambda^0 - \lambda^1\| \\
&\quad + \mu^{-1}\|K\|(\|K\|\|x^1 - x^0\| + \|B\|\lambda^0 - \lambda^1\| + \mu\|z^1 - z^0\|)]^2 \\
&\quad + [\|K\|\|x^1 - x^0\| + \|B\|\lambda^0 - \lambda^1\| + 2\mu\|z^1 - z^0\|]^2 \\
&\quad + [\|A\|\|x^1 - x^0\| + \mu^{-1}\|B\|(\|K\|\|x^1 - x^0\| + \|B\|\lambda^0 - \lambda^1\| + \mu\|z^1 - z^0\|)]^2 \\
&\leq \frac{3}{\mu^2}\left[(L_g\mu + \|K\|^2)^2 + (\|A\|\mu + \|K\|\|B\|)^2 + \mu^2\|K\|^2\right]\|x^1 - x^0\|^2 \\
&\quad + 3(4\mu^2 + \|K\|^2 + \|B\|^2)\|z^1 - z^0\|^2 \\
&\quad + \frac{3}{\mu^2}\left[(\mu\|A\| + \|K\|\|B\|)^2 + \|B\|^4 + \mu^2\|B\|^2\right]\|\lambda^1 - \lambda^0\|^2 \\
&\leq \frac{\gamma^2}{\mu^2}[\|x^1 - x^0\|^2 + \|z^1 - z^0\|^2 + \|\lambda^1 - \lambda^0\|^2],
\end{aligned}
$$

which implies the estimate (2.7). The proof is completed. ∎

Define $L_\theta = \gamma/\mu$. Then, θ is continuously differentiable and $\nabla\theta$ is L_θ-Lipschitz continuous when Assumption 2.1 is satisfied. Next, we establish the optimality conditions of problem (1.1). We define the stationary point of problem (1.1), which is similar to the KKT point for constrained optimization problems.

Definition 2.2. *We say that* $(\overline{x},\overline{y})$ *is a stationary point of problem* (1.1) *if there exists a vector* $\overline{\lambda}\in\Re^q$ *such that*

(2.13)
$$0\in K\overline{y}+A^T\overline{\lambda}+\nabla g(\overline{x}) + \partial\varphi(\overline{x}),\ 0\in K^T\overline{x} + B^T\overline{\lambda} - \nabla h(\overline{y}) - \partial\psi(\overline{y}),\ A\overline{x} + B\overline{y} + c = 0.$$

The following proposition shows the equivalence condition of the stationary point of the minimax problem (1.1).

Proposition 2.3. *Let Assumption* 2.1 *be satisfied. Then*

$$0\in\nabla\theta(\overline{x},\overline{y},\overline{\lambda}) + \partial\varphi(\overline{x})\times\{0\}\times\{0\}$$

if and only if $(\overline{x},\overline{y})$ *is a stationary point of problem* (1.1).

Proof. If $0\in\nabla\theta(\overline{x},\overline{y},\overline{\lambda}) + \partial\varphi(\overline{x})\times\{0\}\times\{0\}$, we have from (2.6) that

$$
\begin{aligned}
&0\in\nabla g(\overline{x}) + A^T\overline{\lambda} + Ky_*(\overline{x},\overline{y},\overline{\lambda}) + \partial\varphi(\overline{x}),\\
&0 = \overline{y} - y_*(\overline{x},\overline{y},\overline{\lambda}),\\
&0 = A\overline{x} + c + By_*(\overline{x},\overline{y},\overline{\lambda}).
\end{aligned}
$$

This implies $\overline{y} = y_*(\overline{x},\overline{y},\overline{\lambda})$ and in turn

$$0\in\partial_y Q(\overline{x},\overline{y},\overline{y},\overline{\lambda}),\ 0\in\nabla g(\overline{x}) + A^T\overline{\lambda} + K\overline{y} + \partial\varphi(\overline{x}),\ 0 = A\overline{x} + c + B\overline{y},$$

which implies that $(\overline{x},\overline{y})$ is a stationary point of problem (1.1).

On the other hand, let $(\overline{x},\overline{y},\overline{\lambda})$ satisfy (2.13), or equivalently

$$0\in K\overline{y} + A^T\overline{\lambda} + \partial\varphi_g(\overline{x}),\ 0\in K^T\overline{x} + B^T\overline{\lambda} - \partial\psi_h(\overline{y}),\ A\overline{x} + B\overline{y} + c = 0. \tag{2.14}$$

Then, from $0\in K^T\overline{x} + B^T\overline{\lambda} - \partial\psi_h(\overline{y})$, we have

$$0\in\nabla_y[Q(\overline{x},\overline{y},y,\overline{\lambda}) + \psi(y)]|_{y=\overline{y}} - \psi(\overline{y}),$$

which implies that $\overline{y} = \operatorname{argmax}\left\{Q(\overline{x},\overline{y},y,\overline{\lambda})\right\}$, namely, $\overline{y} = y_*(\overline{x},\overline{y},\overline{\lambda})$, and in this case,

$$\nabla\theta(\overline{x},\overline{y},\overline{\lambda}) = \begin{pmatrix} \nabla g(\overline{x}) + A^T\overline{\lambda} + Ky_*(\overline{x},\overline{y},\overline{\lambda}) \\ \overline{y} - y_*(\overline{x},\overline{y},\overline{\lambda}) \\ A\overline{x} + c + By_*(\overline{x},\overline{y},\overline{\lambda}) \end{pmatrix} = \begin{pmatrix} \nabla g(\overline{x}) + A^T\overline{\lambda} + K\overline{y} \\ 0 \\ A\overline{x} + c + B\overline{y} \end{pmatrix},$$

which implies that $0 \in \nabla\theta(\overline{x},\overline{y},\overline{\lambda}) + \partial\varphi(\overline{x}) \times \{0\} \times \{0\}$ from (2.14). The proof is completed. ∎

Let

$$f(x,y,\lambda) = g(x) + x^TKy - h(y) + \langle\lambda, Ax + By + c\rangle. \tag{2.15}$$

Then, under Assumption 2.1, problem (1.1) is reformulated as the following unconstrained nonsmooth minimax optimization problem:

$$\min_{x,\lambda}\max_{y} \mathcal{L}(x,y,\lambda) = \varphi(x) + f(x,y,\lambda) - \psi(y). \tag{2.16}$$

Obviously, we have that $(\overline{x},\overline{y},\overline{\lambda})$ is a stationary point of problem (2.16) if and only if $(\overline{x},\overline{y},\overline{\lambda})$ satisfies (2.13).

As $\partial\varphi$ and $\partial\psi$ are hard to portray, the proximal mapping is used to characterize the properties of the stationary point of problem (1.1). Hence, let $L > 0$ and define

$$\begin{aligned} G_L^{f,\varphi}(x,y,\lambda) &= L(x - \operatorname{prox}_{L^{-1}\varphi}(x - L^{-1}\nabla_x f(x,y,\lambda))), \\ G_L^{f,0}(x,y,\lambda) &= L(\lambda - \operatorname{prox}_{L^{-1}0}(\lambda - L^{-1}\nabla_\lambda f(x,y,\lambda))), \\ G_L^{f,\psi}(x,y,\lambda) &= L(y - \operatorname{prox}_{L^{-1}\psi}(y + L^{-1}\nabla_y f(x,y,\lambda))). \end{aligned}$$

Then we have

$$\begin{aligned} G_L^{f,\varphi}(x,y,\lambda) &= L\left(x - \operatorname{prox}_{L^{-1}\varphi}\left(x - L^{-1}\left(\nabla g(x) + Ky + A^T\lambda\right)\right)\right), \\ G_L^{f,0}(x,y,\lambda) &= Ax + By + c, \\ G_L^{f,\psi}(x,y,\lambda) &= L\left(y - \operatorname{prox}_{L^{-1}\psi}\left(y + L^{-1}\left(-\nabla h(y) + K^Tx + B^T\lambda\right)\right)\right). \end{aligned}$$

It is not hard to verify the following result.

LEMMA 2.4. *For $L_1 > 0$, $L_2 > 0$, and $L_3 > 0$, $(\overline{x},\overline{y})$ is a stationary point of problem* (1.1) *if and only if there exists $\overline{\lambda} \in \Re^q$ such that*

$$G_{L_1}^{f,\varphi}(\overline{x},\overline{y},\overline{\lambda}) = 0,\ G_{L_2}^{f,\psi}(\overline{x},\overline{y},\overline{\lambda}) = 0,\ \text{and}\ G_{L_3}^{f,0}(\overline{x},\overline{y},\overline{\lambda}) = 0.$$

DEFINITION 2.5. *For $\epsilon > 0$, we say that (x,y) is an ϵ-stationary point of problem* (1.1) *if and only if there exists a vector $\lambda \in \Re^q$ such that*

$$\left\|G_{L_1}^{f,\varphi}(x,y,\lambda)\right\| \le \epsilon,\ \left\|G_{L_2}^{f,\psi}(x,y,\lambda)\right\| \le \epsilon,\ \text{and}\ Ax + By + c = 0$$

for some positive constants L_1 and L_2.

Let $f_\mu(x,z,y,\lambda) = f(x,y,\lambda) + \mu\|y - z\|^2/2$ and $Q_0(x,z,y,\lambda) = x^TKy - h(y) + \langle\lambda, Ax + By + c\rangle - \frac{\mu}{2}\|y - z\|^2$. It follows from the definitions of f_μ that

$$\begin{aligned} G_L^{f,\varphi}(x,y,\lambda) &= G_L^{f_\mu,\varphi}(x,z,y,\lambda) = G_L^{Q_0,\varphi}(x,z,y,\lambda), \\ G_L^{f,0}(x,y,\lambda) &= G_L^{f_\mu,0}(x,z,y,\lambda) = G_L^{Q_0,0}(x,z,y,\lambda), \\ G_L^{f,\psi}(x,y,\lambda) &= G_L^{f_\mu,\psi}(x,y,y,\lambda) = G_L^{Q_0,\psi}(x,y,y,\lambda). \end{aligned} \tag{2.17}$$

3. An alternating coordinate method. Algorithm 3.1 is trying to get an approximate stationary point of problem (1.1), whose main idea is to construct an alternating coordinate ascent descent method for solving problem (2.16).

Algorithm 3.1 is more like an algorithmic framework, where the technique of solving y is not specified in (3.1). The convergence of Algorithm 3.1 only depends on whether the subproblem can be solved accurately enough. For simplicity, we denote $y_*(t) := y_*(x^t, z^t, \lambda^t) \in \operatorname{argmax}_{y'} Q(x^t, z^t, y', \lambda^t)$. Then $G_L^{Q_0,\psi}(x^t, z^t, y, \lambda^t) = 0$ for some constant $L > 0$ if and only if there exists a unique $y_*(t)$ such that $y = y_*(t)$, which implies that $G_L^{f,\psi}(x^t, y, \lambda^t) = 0$ for some constant $L > 0$ if and only if there exists a unique $y_*(t)$ such that $y = y_*(t) = z^t$. To give the convergence properties of Algorithm 3.1, we need the following assumption to ensure the finite optimal value of the minimax problem (2.16), which is widely used in [11, 42, 43].

Assumption 3.1. Suppose that φ_g, ψ_h, and $A,\ B,\ c$ satisfy

$$\inf_{x,z,\lambda} \theta_\varphi(x, z, \lambda) > -\infty.$$

For studying problem (2.16), Assumption 3.1 is reasonable when we try to find a local minimax point of unconstrained minimax optimization problem (2.16) in the sense of Jin, Netrapalli, and Jordan [36].

Assumption 3.2. Suppose that there exists a constant $\rho_0 > 0$ such that the following error bound condition holds:

$$\|y - y_*(t)\| \leq \rho_0 \|G_1^{Q_0,\psi}(x^t, z^t, y, \lambda^t)\| \quad \forall y \in \mathbf{B}\left(y_*(t), \|y^t - y_*(t)\|\right).$$

Assumption 3.2 is a conventional error bound condition for $\max_y Q(x^t, z^t, y, \lambda^t)$. In fact, Assumption 3.2 holds trivially if h is strongly concave and $\psi(y) \equiv 0$. For given $(x^t, z^t, \lambda^t) \in \operatorname{dom}\varphi \times \Re^m \times \Re^q$, let $y_*(t)$ satisfy $G_L^{Q_0,\psi}(x^t, z^t, y_*(t), \lambda^t) = 0$ and

$$\xi^t = \begin{bmatrix} \nabla_x f_\mu(x^t, z^t, y^{t+1}, \lambda^t) \\ \nabla_z f_\mu(x^t, z^t, y^{t+1}, \lambda^t) \\ \nabla_\lambda f_\mu(x^t, z^t, y^{t+1}, \lambda^t) \end{bmatrix} = \begin{bmatrix} \nabla g(x^t) + K y^{t+1} + A^T \lambda^t \\ \mu(-z^t + y^{t+1}) \\ A x^t + B y^{t+1} + c \end{bmatrix}. \tag{3.2}$$

PROPOSITION 3.1. *Let Assumptions* 2.1, 3.1, *and* 3.2 *be satisfied. Let* $\alpha_x \in (0, 1/L_\theta)$ *with* $L_\theta = \gamma/\mu$ *and the sequence* $\{(x^t, z^t, y^t, \lambda^t) : t = 0, 1, \ldots,\}$ *be generated*

Algorithm 3.1. Alternating coordinate ascent descent method.

Input $(x^0, z^0, y^0, \lambda^0) \in \Re^n \times \Re^m \times \Re^\times \Re^q$, $\alpha_x > 0$, $\varepsilon_t > 0$ for $t \in \mathbf{N}$

for $t = 0, 1, 2, \ldots,$ **do**

Find y^{t+1} such that

$$\|G_1^{Q_0,\psi}(x^t, z^t, y^{t+1}, \lambda^t)\| \leq \varepsilon_t; \tag{3.1}$$

Calculate

$$\begin{aligned} x^{t+1} &= \operatorname{prox}_{\alpha_x \varphi}\left[x^t - \alpha_x\left(\nabla g(x^t) + K y^{t+1} + A^T \lambda^t\right)\right], \\ z^{t+1} &= (1 + \alpha_x \mu) z^t - \alpha_x \mu y^{t+1}, \\ \lambda^{t+1} &= \lambda^t - \alpha_x [A x^t + B y^{t+1} + c]. \end{aligned}$$

end for

return $(x^{t+1}, y^{t+1}, \lambda^{t+1})$ for $t = 0, 1, 2, \ldots$

---

by Algorithm 3.1*. Suppose that there exists a sequence* $y_*(t) \in \operatorname{argmax}_{y'} Q(x^t, z^t, y', \lambda^t)$ *such that* $\|y^{t+1} - y_*(t)\| \le \|y^t - y_*(t)\|$ *and*

$$\sum_{t=0}^{\infty} \varepsilon_t^2 < \infty. \tag{3.3}$$

Then for $t = 0, 1, \ldots$ *the following hold.*

(i) *The following series is convergent:*

$$\sum_{t=0}^{\infty} \left\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\right\|^2 < \infty. \tag{3.4}$$

(ii) *For* $\sigma(x, z, \lambda) = \varphi(x)$*, the following sequences are convergent:*

$$G^{\theta,\sigma}_{\alpha_x^{-1}}(x^t, z^t, y_*(t), \lambda^t) \to 0,\ G^{Q_0,\psi}_1(x^t, z^t, y^t, \lambda^t) \to 0. \tag{3.5}$$

(iii) *Let* $(\overline{x}, \overline{z}, \overline{y}, \overline{\lambda})$ *be any accumulation point of* $\{(x^t, z^t, y^t, \lambda^t)\}$*. Then* $(\overline{x}, \overline{y})$ *is a stationary point of problem* (1.1) *with Lagrange multiplier* $\overline{\lambda}$*.*

Proof. For ξ^t defined by (3.2), $(x^{t+1}, z^{t+1}, \lambda^{t+1})$ can be expressed as

$$(x^{t+1}, z^{t+1}, \lambda^{t+1}) = \operatorname{prox}_{\alpha_x \sigma}((x^t, z^t, \lambda^t) - \alpha_x \xi^t), \tag{3.6}$$

where $\sigma(x, z, \lambda) = \varphi(x)$. Let $y_*(t)$ satisfy $G^{f,\psi}_L(x^t, z^t, y_*(t), \lambda^t) = 0$ and $\xi^t_* = \nabla\theta(x^t, z^t, \lambda^t)$. From Proposition 2.1 and applying Lemma A.2 with $h(x, z, \lambda) = f(x, z, y_*(x, z, \lambda), \lambda)$ and $\sigma(x, z, \lambda) = \varphi(x)$, we obtain

$$\begin{aligned}
&\varphi(x^{t+1}) + f_\mu(x^{t+1}, z^{t+1}, y_*(t+1), \lambda^{t+1}) - \psi(y_*(t+1)) \le \varphi(x^t) + f_\mu(x^t, z^t, y_*(t), \lambda^t) \\
&\quad - \ \psi(y_*(t))\frac{\alpha_x^{-1} - L_\theta}{2}\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\|^2 \\
&\quad + \ \langle \xi^t_* - \xi^t, (x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\rangle.
\end{aligned} \tag{3.7}$$

Using the simple inequality $\langle a, b\rangle \le \|a\|^2/2 + \|b\|^2/2$ for vectors a and b, we obtain

$$\begin{aligned}
&\langle \xi^t_* - \xi^t, (x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\rangle \\
&\quad \le \frac{\alpha_x^{-1} - L_\theta}{4}\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\|^2 + \frac{\|\xi^t_* - \xi^t\|^2}{(\alpha_x^{-1} - L_\theta)}.
\end{aligned} \tag{3.8}$$

We now analyze the term $\|\xi^t_* - \xi^t\|^2$. Noting that $\nabla_x Q(x^t, z, \cdot, \lambda^t)$, $\nabla_z Q(x^t, z, \cdot, \lambda^t)$ and $\nabla_\lambda Q(x^t, z, \cdot, \lambda^t)$ are $\|K\|$-Lipschitz continuous, μ-Lipschitz continuous, and $\|B\|$-Lipschitz continuous, respectively, it follows from $\xi^t = \nabla_{x,z,\lambda}\ f_\mu(x^t, z^t, y^{t+1}, \lambda^t) = \nabla g(x^t) + \nabla_{x,z,\lambda} Q(x^t, z^t, y^{t+1},\ \lambda^t)$ and $\xi^t_* = \nabla_{x,z,\lambda}\ f_\mu(x^t, z^t, y_*(t), \lambda^t) = \nabla g(x^t) + \nabla_{x,z,\lambda} Q(x^t, z^t, y_*(t), \lambda^t)$ that

$$\begin{aligned}
\|\xi^t - \xi^t_*\|^2 &\le \|\nabla_x Q(x^t, z^t, y^{t+1}, \lambda^t) - \nabla_x Q(x^t, z^t, y_*(t), \lambda^t)\|^2 \\
&\quad + \|\nabla_z Q(x^t, z^t, y^{t+1}, \lambda^t) - \nabla_z Q(x^t, z^t, y_*(t), \lambda^t)\|^2 \\
&\quad + \|\nabla_\lambda Q(x^t, z^t, y^{t+1}, \lambda^t) - \nabla_\lambda Q(x^t, z^t, y_*(t), \lambda^t)\|^2 \\
&\le [\|K\|^2 + \mu^2 + \|B\|^2]\|y^{t+1} - y_*(t)\|^2.
\end{aligned} \tag{3.9}$$

Since $\|y^{t+1} - y_*(t)\| \le \|y^t - y_*(t)\|$, from the error bound condition (3.2), this inequality implies

(3.10)
$$\|\xi^t - \xi^t_*\|^2 \le [\|K\|^2 + \mu^2 + \|B\|^2]\|G_1^{Q_0,\psi}(x^t, z^t, y, \lambda^t)\|^2 \le [\|K\|^2 + \mu^2 + \|B\|^2]\varepsilon_t^2.$$

Then, from (3.7), (3.8), and (3.10), we obtain the following inequality:

(3.11)
$$\frac{1-\alpha_x L_\theta}{4\alpha_x}\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\|^2 \le \theta_\varphi(x^t, z^t, \lambda^t) - \theta_\varphi(x^{t+1}, z^{t+1}, \lambda^{t+1}) + \frac{\|K\|^2 + \mu^2 + \|B\|^2}{(\alpha_x^{-1} - L_\theta)}\varepsilon_t^2.$$

For $T > 1$, summing (3.11) over $t = 0, 1, \ldots, T-1$, we obtain from Assumption 3.1 that

$$\sum_{t=0}^{T-1}\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\|^2 \le \frac{4\alpha_x}{1-\alpha_x L_\theta}[\theta_\varphi(x^0, z^0, \lambda^0) - \inf\theta_\varphi] + \frac{4\alpha_x(\|K\|^2 + \mu^2 + \|B\|^2)}{(1-\alpha_x L_\theta)^2}\sum_{t=0}^{T-1}\varepsilon_t^2.$$

Therefore, we obtain (3.4) of (i) from condition (3.3).

Now we turn to the proof of (ii). Noting that $y_*(t+1)$ is the unique solution of $G_L^{Q_0,\psi}(x^{t+1},\ z^{t+1}, y, \lambda^{t+1}) = 0$, it follows from (3.6) and the definition of $G^{\theta,\sigma}_{\alpha_x^{-1}}(x^{t+1}, z^{t+1}, y_*(t+1), \lambda^{t+1})$ that

$$\begin{aligned}
&\left\|G^{\theta,\sigma}_{\alpha_x^{-1}}(x^{t+1}, z^{t+1}, y_*(t+1), \lambda^{t+1})\right\| \\
&= \alpha_x^{-1}\left\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - \mathrm{prox}_{\alpha_x\sigma}((x^{t+1}, z^{t+1}, \lambda^{t+1}) - \alpha_x\xi_*^{t+1})\right\| \\
&= \alpha_x^{-1}\left\|\mathrm{prox}_{\alpha_x\sigma}((x^t, z^t, \lambda^t) - \alpha_x\xi^t) - \mathrm{prox}_{\alpha_x\sigma}((x^{t+1}, z^{t+1}, \lambda^{t+1}) - \alpha_x\xi_*^{t+1})\right\| \\
&\le \alpha_x^{-1}\left\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\right\| + \left\|\xi^t - \xi_*^{t+1}\right\| \\
&\le \alpha_x^{-1}\left\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\right\| + \left\|\xi^t - \xi_*^t\right\| + \left\|\xi_*^t - \xi_*^{t+1}\right\|.
\end{aligned}$$

Noting that

$$\begin{aligned}
\left\|\xi_*^t - \xi_*^{t+1}\right\| &\le \left\|\nabla\theta(x^t, z^t, \lambda^t) - \nabla\theta(x^{t+1}, z^{t+1}, \lambda^{t+1})\right\| \\
&\le L_\theta\left\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\right\|,
\end{aligned}$$

it follows from (3.10) that

$$\left\|G^{Q_0,\varphi}_{\alpha_x^{-1}}(x^{t+1}, z^{t+1}, y_*(t+1), \lambda^{t+1})\right\| \le (\alpha_x^{-1} + L_\theta)\left\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\right\| + \sqrt{\|K\|^2 + \mu^2 + \|B\|^2}\varepsilon_t.$$

Therefore, we obtain $\|G^{Q_0,\varphi}_{\alpha_x^{-1}}(x^{t+1}, z^{t+1}, y_*(t+1), \lambda^{t+1})\| \to 0$ from (3.3) and (3.4). From the definition of $G_1^{Q_0,\psi}(x, z, y, \lambda)$, we have

$$\begin{aligned}
&\left\|G_1^{Q_0,\psi}(x^{t+1}, z^{t+1}, y^{t+1}, \lambda^{t+1})\right\| \\
&\quad\le \left\|G_1^{Q_0,\psi}(x^{t+1}, z^{t+1}, y^{t+1}, \lambda^{t+1}) - G_1^{Q_0,\psi}(x^t, z^{t+1}, y^{t+1}, \lambda^t)\right\| \\
&\qquad + \left\|G_1^{Q_0,\psi}(x^t, z^t, y^{t+1}, \lambda^t)\right\|
\end{aligned}$$

$$
\begin{aligned}
&= \left\|G_1^{f,\psi}(x^t, z^t, y^{t+1}, \lambda^t)\right\| + \|\mathrm{prox}_\psi(y^{t+1} - \nabla h(y^{t+1}) + K^T x^{t+1} + B^T \lambda^{t+1} \\
&\quad -\mu(y^{t+1} - z^{t+1})) - \mathrm{prox}_\psi(y^{t+1} - \nabla h(y^{t+1}) + K^T x^t + B^T \lambda^t - \mu(y^{t+1} - z^t)))\| \\
&\le \varepsilon_t + \|K\| \left\|x^{t+1} - x^t\right\| + \|B\| \left\|\lambda^{t+1} - \lambda^t\right\| + \mu \left\|z^{t+1} - z^t\right\|,
\end{aligned}
$$

which converges to zero when $t \to \infty$ from (3.3) and (3.4).

Let $(\overline{x}, \overline{z}, \overline{y}, \overline{\lambda})$ be an accumulation point of $\{(x^t, z^t, y^t, \lambda^t)\}$. From the Lipschitz continuity of $G_{\alpha_x^{-1}}^{Q_0,\varphi}$ and the Lipschitz continuity of $G_1^{Q_0,\psi}$, we obtain

$$
G_{\alpha_x^{-1}}^{\theta,\sigma}(\overline{x}, \overline{z}, y_*(\overline{x}, \overline{z}, \overline{\lambda}), \overline{\lambda}) = 0, \quad G_1^{Q_0,\psi}(\overline{x}, \overline{z}, \overline{y}, \overline{\lambda}) = 0,
$$

which implies that

$$
G_{\alpha_x^{-1}}^{f,\ \varphi}(\overline{x}, y_*(\overline{x}, \overline{z}, \overline{\lambda}), \overline{\lambda}) = 0,\ \overline{z} = y_*(\overline{x}, \overline{z}, \overline{\lambda}),\ \text{and } G_{\alpha_x^{-1}}^{f,0}(\overline{x}, y_*(\overline{x}, \overline{z}, \overline{\lambda}), \overline{\lambda}) = 0.
$$

Since $y_*(\overline{x}, \overline{z}, \overline{\lambda}) \in \mathrm{argmax}_{y'} Q(\overline{x}, \overline{z}, y', \overline{\lambda})$ is the unique solution of $G_1^{Q_0,\psi}(\overline{x}, \overline{z}, y, \overline{\lambda}) = 0$, we have $\overline{y} = y_*(\overline{x}, \overline{z}, \overline{\lambda})$, which implies that $\overline{y} = \overline{z}$. Therefore, we obtain from (2.17) that

$$
G_{\alpha_x^{-1}}^{f,\varphi}(\overline{x}, \overline{y}, \overline{\lambda}) = 0, \quad G_{\alpha_x^{-1}}^{f,0}(\overline{x}, \overline{y}, \overline{\lambda}) = 0, \quad \text{and} \quad G_1^{f,\psi}(\overline{x}, \overline{y}, \overline{\lambda}) = G_1^{Q_0,\psi}(\overline{x}, \overline{y}, \overline{y}, \overline{\lambda}) = 0.
$$

From Lemma 2.4, $(\overline{x}, \overline{y})$ is a stationary point of problem (1.1). ∎

Remark 3.2. The condition $\|y^{t+1} - y_*(t)\| \le \|y^t - y_*(t)\|$ required by Proposition 3.1 can easily be guaranteed; for example, we may use the gradient ascent method with a constant stepsize.

4. A proximal gradient method.

The following algorithm is based on getting an approximate stationary point of problem (2.16), which is a proximal gradient multistep ascent descent method.

In Algorithm 4.1, for fixed point (x^t, z^t, λ^t), a scheme for solving the approximate optimal solution of y by the proximal gradient method is given. Similar to section 3, we denote $y_*(t) := y_*(x^t, z^t, \lambda^t) \in \mathrm{argmax}_{y'} Q(x^t, z^t, y', \lambda^t)$, which is the unique solution of $G_L^{Q_0,\psi}(x^t, z^t, y, \lambda^t) = 0$ for some constant $L > 0$.

We have the following proposition, whose proof is based on Theorem 10.29 of [6].

Algorithm 4.1. Proximal gradient multistep ascent descent method.

Input $(x^0, z^0, y^0, \lambda^0) \in \Re^n \times \Re^m \times \Re^\times \Re^q$, $\alpha_x > 0$, $\alpha_y(t) > 0$ for $t \in \mathbf{N}$, positive integers $T > 1$ and $N_t > 1$ for $t = 0, 1, 2, \dots, T$

for $t = 0, 1, 2, \dots, T-1$, **do**

Set $y^{[0]}(t) = y^t$

for $k = 0, 1, 2, \dots, N_t - 1$, **do**

$$
\begin{aligned}
y^{[k+1]}(t) = \mathrm{prox}_{\alpha_y(t)\psi}[&y^{[k]}(t) + \alpha_y(t)(-\nabla h(y^{[k]}(t)) + K^T x^t + B^T \lambda^t \\
&-\mu(y^{[k]}(t) - z^t))]
\end{aligned}
$$

end for

Set $x^{t+1} = \mathrm{prox}_{\alpha_x \varphi}\left[x^t - \alpha_x \left(\nabla g(x^t) + K y^{t+1} + A^T \lambda^t\right)\right]$.

$z^{t+1} = (1 + \alpha_x \mu) z^t - \alpha_x \mu y^{t+1}$.

$\lambda^{t+1} = \lambda^t - \alpha_x [A x^t + B y^{t+1} + c]$.

end for

return $(x^{t+1}, y^{t+1}, \lambda^{t+1})$ for $t = 0, 1, 2, \dots, T-1$.

PROPOSITION 4.1. *Let Assumption* 2.1 *be satisfied and let* $\alpha_y(t) \in (0, 1/L_h)$. *Consider the sequence* $\{y^{[k]}(t) : k = 0, 1, \dots, N_t\}$ *generated by Algorithm* 4.1, *where* $t = 0, 1, \dots, T$. *Then we have, for* $k = 0, \dots, N_t - 1$,

(a) $\|y^{[k+1]}(t) - y_*(t)\|^2 \le (1 - \mu\alpha_y(t))\|y^{[k]}(t) - y_*(t)\|^2$;
(b) $\|y^{[k+1]}(t) - y_*(t)\|^2 \le (1 - \mu\alpha_y(t))^{k+1}\|y^t - y_*(t)\|^2$;
(c) $\theta_0(x^t, \lambda^t) - Q(x^t, y^{[k+1]}(t), \lambda^t) \le (2\alpha_y(t))^{-1}(1 - \mu\alpha_y(t))^k\|y^t - y_*(t)\|^2$.

Proof. From the assumptions in this proposition, the function $q^t(y) := -Q_0(x^t, z^t, y, \lambda^t)$ is L_h-smooth and the function $y \to -Q(x^t, z^t, y, \lambda^t) = q^t(y) + \psi(y)$ is μ-strongly convex. Note that for $t = 0, 1, \dots, T-1$, the sequence $\{y^{[k]}(t) : k = 0, 1, \dots, N\}$ satisfies

$$y^{[k+1]}(t) = \mathrm{prox}_{\alpha_y(t)\psi}\left[y^{[k]}(t) - \alpha_y(t)\nabla q^t(y^{[k]}(t))\right].$$

From Lemma A.3, we have for $\alpha_y(t) \in (0, 1/L_h)$,

(4.1)
$$\begin{aligned}
&Q(x^t, z^t, y^{[k+1]}(t), \lambda^t) - \theta_0(x^t, z^t, \lambda^t) \\
&\quad \ge \frac{1}{2\alpha_y(t)}\left\|y_*(t) - y^{[k+1]}(t)\right\|^2 - \frac{1}{2\alpha_y(t)}\left\|y_*(t) - y^{[k]}(t)\right\|^2 + \frac{\mu}{2}\left\|y_*(t) - y^{[k]}(t)\right\|^2.
\end{aligned}$$

Since $\theta_0(x^t, z^t, \lambda^t) = \max_y Q(x^t, z^t, y, \lambda^t)$, $Q(x^t, z^t, y^{[k+1]}(t), \lambda^t) - \theta_0(x^t, z^t, \lambda^t) \le 0$, we have from the above inequality that

$$\left\|y_*(t) - y^{[k+1]}(t)\right\|^2 \le (1 - \mu\alpha_y(t))\left\|y_*(t) - y^{[k]}(t)\right\|^2,$$

which establishes part (a). Part (b) follows from (a) immediately. To prove part (c), by (4.1), we have

$$\begin{aligned}
&\theta_0(x^t, z^t, \lambda^t) - Q\left(x^t, z^t, y^{[k+1]}(t), \lambda^t\right) \\
&\quad \le \frac{\alpha_y^{-1}(t) - \mu}{2}\left\|y_*(t) - y^{[k]}(t)\right\|^2 - \frac{1}{2\alpha_y(t)}\left\|y_*(t) - y^{[k+1]}(t)\right\|^2 \\
&\quad \le \frac{\alpha_y^{-1}(t) - \mu}{2}\left\|y_*(t) - y^{[k]}(t)\right\|^2 \\
&\quad = (2\alpha_y(t))^{-1}(1 - \mu\alpha_y(t))^k\left\|y^t - y_*(t)\right\|^2,
\end{aligned}$$

where part (b) was used in the last inequality. This completes our proof. ∎

THEOREM 4.2. *Let Assumption* 2.1 *be satisfied and let* $\alpha_y(t) \in (0, 1/L_h)$. *Consider the sequence* $\{y^{[k]}(t) : k = 0, 1, \dots, N_t\}$ *generated by Algorithm* 4.1, *and* $y^{t+1} = y^{[N_t]}(t)$, *where* $t = 0, 1, \dots, T$. *Let* $y_*(t) := y_*(x^t, z^t, \lambda^t) \in \mathrm{argmax}_{y'}\, Q(x^t, z^t, y', \lambda^t)$ *be the unique solution of* $G^{Q_0,\psi}_{\alpha_y(t)}(x^t, z^t, y, \lambda^t) = 0$. *Then one has for* $t = 0, 1, \dots, T-1$ *that*

(4.2)
$$\begin{aligned}
&\left\|y^{t+1} - y_*(t)\right\|^2 \le (1 - \mu\alpha_y(t))^{N_t}\|y^t - y_*(t)\|^2, \\
&\left\|\nabla_x Q(x^t, z^t, y^{t+1}, \lambda^t) - \nabla_x Q(x^t, z^t, y_*(t), \lambda^t)\right\|^2 \le \|K\|^2(1 - \mu\alpha_y(t))^{N_t}\left\|y^t - y_*(t)\right\|^2, \\
&\left\|\nabla_z Q(x^t, z^t, y^{t+1}, \lambda^t) - \nabla_x Q(x^t, z^t, y_*(t), \lambda^t)\right\|^2 \le \mu^2(1 - \mu\alpha_y(t))^{N_t}\left\|y^t - y_*(t)\right\|^2, \\
&\left\|\nabla_\lambda Q(x^t, z^t, y^{t+1}, \lambda^t) - \nabla_\lambda Q(x^t, z^t, y_*(t), \lambda^t)\right\|^2 \le \|B\|^2(1 - \mu\alpha_y(t))^{N_t}\|y^t - y_*(t)\|^2, \\
&\left\|G^{Q_0,\psi}_{\alpha_y^{-1}(t)}(x^t, z^t, y^{t+1}, \lambda^t)\right\|^2 \le 9\alpha_y^{-2}(t)(1 - \mu\alpha_y(t))^{N_t}\left\|y^t - y_*(t)\right\|^2
\end{aligned}$$

and

(4.3)

$$\left\|y^{t+1}-y_*(t)\right\|^2 \le \frac{2}{\mu}(1-\mu\alpha_y(t))^{N_t}[\theta_0(x^t,z^t,\lambda^t)-Q(x^t,z^t,y^t,\lambda^t)],$$

$$\|\nabla_x Q(x^t,z^t,y^{t+1},\lambda^t)-\nabla_x Q(x^t,z^t,y_*(t),\lambda^t)\|^2 \le \frac{2\|K\|^2}{\mu}(1-\mu\alpha_y(t))^{N_t}(\theta_0(x^t,z^t,\lambda^t) - Q(x^t,z^t,y^t,\lambda^t)),$$

$$\|\nabla_z Q(x^t,z^t,y^{t+1},\lambda^t)-\nabla_z Q(x^t,z^t,y_*(t),\lambda^t)\|^2 \le 2\mu(1-\mu\alpha_y(t))^{N_t}(\theta_0(x^t,z^t,\lambda^t) - Q(x^t,z^t,y^t,\lambda^t)),$$

$$\|\nabla_\lambda Q(x^t,z^t,y^{t+1},\lambda^t)-\nabla_\lambda Q(x^t,z^t,y_*(t),\lambda^t)\|^2 \le \frac{2\|B\|^2}{\mu}(1-\mu\alpha_y(t))^{N_t}(\theta_0(x^t,z^t,\lambda^t) - Q(x^t,z^t,y^t,\lambda^t)),$$

$$\|G^{Q_0,\psi}_{\alpha_y^{-1}(t)}(x^t,z^t,y^{t+1},\lambda^t)\|^2 \le \frac{18\alpha_y^{-2}(t)}{\mu}(1-\mu\alpha_y(t))^{N_t}(\theta_0(x^t,z^t,\lambda^t)-Q(x^t,z^t,y^t,\lambda^t)).$$

Proof. It follows from Proposition 4.1 (b) that

$$\left\|y^{t+1}(t)-y_*(t)\right\|^2 \le (1-\mu\alpha_y(t))^{N_t}\|y^t-y_*(t)\|^2, \tag{4.4}$$

which is just the first inequality in (4.2). By $\nabla_x Q(x^t,z^t,y_*(t),\lambda^t)=Ky_*(t)+A^T\lambda^t$, $\nabla_z Q(x^t,z^t,\, y_*(t),\lambda^t)=\mu(y_*(t)-z^t)$, and $\nabla_\lambda Q(x^t,z^t,y_*(t),\lambda^t)=Ax^t+By_*(t)+c$, we have that $\nabla_x Q(x^t,z^t,\cdot,\lambda^t)$, $\nabla_z Q(x^t,z^t,\cdot,\lambda^t)$, and $\nabla_\lambda Q(x^t,z^t,\cdot,\lambda^t)$ are $\|K\|$-Lipschitz continuous, μ-Lipschitz continuous, and $\|B\|$-Lipschitz continuous, respectively, which yields that

$$\begin{aligned}
\left\|\nabla_x Q(x^t,z^t,y^{t+1},\lambda^t)-\nabla_x Q(x^t,z^t,y_*(t),\lambda^t)\right\|^2 &\le \|K\|^2\left\|y^{t+1}-y_*(t)\right\|^2,\\
\left\|\nabla_z Q(x^t,z^t,y^{t+1},\lambda^t)-\nabla_x Q(x^t,z^t,y_*(t),\lambda^t)\right\|^2 &\le \mu^2\left\|y^{t+1}-y_*(t)\right\|^2,\\
\left\|\nabla_\lambda Q(x^t,z^t,y^{t+1},\lambda^t)-\nabla_z Q(x^t,z^t,y_*(t),\lambda^t)\right\|^2 &\le \|B\|^2\left\|y^{t+1}-y_*(t)\right\|^2.
\end{aligned}$$

Thus the second, third, and fourth inequalities in (4.2) are obtained from the first inequality in (4.2).

Noting that $G^{Q_0,\psi}_{\alpha_y^{-1}(t)}(x^t,z^t,y_*(t),\lambda^t)=0$, we have from Lemma A.1 in Appendix A that

$$\begin{aligned}
\left\|G^{Q_0,\psi}_{\alpha_y^{-1}(t)}(x^t,z^t,y^{t+1},\lambda^t)\right\|^2 &= \left\|G^{Q_0,\psi}_{\alpha_y^{-1}(t)}(x^t,z^t,y^{t+1},\lambda^t)-G^{Q_0,\psi}_{\alpha_y^{-1}(t)}(x^t,z^t,y_*(t),\lambda^t)\right\|^2\\
&\le 9\alpha_y^{-2}(t)\left\|y^{t+1}-y_*(t)\right\|^2,
\end{aligned}$$

and the last inequality in (4.2) can be obtained from the first inequality in (4.2).

Noting that $y\to Q(x^t,z^t,y,\lambda^t)$ is μ-strongly concave, we have

$$\|y^t-y_*(t)\|^2 \le \frac{2}{\mu}[\theta_0(x^t,z^t,\lambda^t)-Q(x^t,z^t,y^t,\lambda^t)]. \tag{4.5}$$

Thus the inequalities of (4.3) come from (4.5) and (4.4) directly. The proof is completed. ∎

PROPOSITION 4.3. *Let Assumption* 2.1 *be satisfied. Let* $\alpha_y(t)\in(0,1/L_h)$, $\alpha_x\in(0,1/L_\theta)$, *and the sequence* $\{(x^t,z^t,y^t,\lambda^t) : t=0,1,\dots,T\}$ *be generated by Algorithm* 4.1. *Then for* $t=0,\dots,T-1$,

(4.6)
$$\theta_\varphi(x^{t+1}, z^{t+1}, \lambda^{t+1}) \le \theta_\varphi(x^k, z^t, \lambda^k) - \frac{1-\alpha_x L_\theta}{4\alpha_x}\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\|^2 + \frac{\alpha_x(\|K\|^2+\mu^2+\|B\|^2)}{(1-\alpha_x L_\theta)}(1-\mu\alpha_y(t))^{N_t}\|y^t - y_*(t)\|^2$$

and

(4.7)
$$\theta_\varphi(x^{t+1}, z^{t+1}, \lambda^{t+1}) \le \theta_\varphi(x^k, z^t, \lambda^k) - \frac{1-\alpha_x L_\theta}{4\alpha_x}\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\|^2 + \frac{2\alpha_x(\|K\|^2+\mu^2+\|B\|^2)}{\mu(1-\alpha_x L_\theta)}(1-\mu\alpha_y(t))^{N_t}(\theta_0(x^t, z^t, \lambda^t) - Q(x^t, z^t, y^t, \lambda^t)).$$

Proof. From Proposition 4.1, the condition $\|y^{t+1} - y_*(t)\| \le \|y^t - y_*(t)\|$ is guaranteed in Algorithm 4.1. Hence, by (3.7) and (3.8) in Proposition 3.1, we have

$$\theta_\varphi(x^{t+1}, z^{t+1}, \lambda^{t+1}) \le \theta_\varphi(x^t, z^t, \lambda^t) - \frac{\alpha_x^{-1} - L_\theta}{4}\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\|^2 + \frac{\|\xi_*^t - \xi^t\|^2}{(\alpha_x^{-1} - L_\theta)},$$

where $\xi_*^t = \nabla\theta(x^t, z^t, \lambda^t) = \nabla_{x,z,\lambda} f_\mu(x^t, z^t, y_*(t), \lambda^t)$ is as defined in (2.6) and $\xi^t = \nabla_{x,z,\lambda} f_\mu(x^t, z^t, y^{t+1}, \lambda^t)$ is as defined in (3.2). It follows from (3.9) that $\|\xi^t - \xi_*^t\|^2 \le [\|K\|^2 + \mu^2 + \|B\|^2]\|y^{t+1} - y_*(t)\|^2$, which implies that

$$\theta_\varphi(x^{t+1}, z^{t+1}, \lambda^{t+1}) \le \theta_\varphi(x^t, z^t, \lambda^t) - \frac{\alpha_x^{-1} - L_\theta}{4}\|(x^{t+1}, z^{t+1}, \lambda^{t+1}) - (x^t, z^t, \lambda^t)\|^2 + \frac{[\|K\|^2+\mu^2+\|B\|^2]}{(\alpha_x^{-1} - L_\theta)}\|y^{t+1} - y_*(t)\|^2.$$

Hence, the inequalities (4.6) and (4.7) are derived from (4.2) and (4.3) in Theorem 4.2. ∎

Define

$$\chi_0 = (\alpha_x(1 - L_\theta\alpha_x))^{-1}, \quad \chi_1 = (\|K\|^2+\mu^2+\|B\|^2)(1-\alpha_x L_\theta)^{-2},$$

and

(4.8)
$$\delta_t = (1-\mu\alpha_y(t))^{N_t}\|y^t - y_*(t)\|^2, \quad \Delta_t = (1-\mu\alpha_y(t))^{N_t}(\theta_0(x^t, z^t, \lambda^t) - Q(x^t, z^t, y^t, \lambda^t)).$$

PROPOSITION 4.4. *Let Assumptions* 2.1 *and* 3.1 *be satisfied. Let* $\alpha_y(t) \in (0, 1/L_h)$, $\alpha_x \in (0, 1/L_\theta)$, *and the sequence* $\{(x^t, z^t, y^t, \lambda^t) : t = 0, 1, \ldots, T\}$ *be generated by Algorithm* 4.1. *Then there exists an integer* $t \in \{0, 1, \ldots, T-1\}$ *such that*

(4.9)
$$\|G_\mu^{f,\psi}(x^{t+1}, y^{t+1}, \lambda^{t+1})\|^2 \le 8\chi_0\frac{\mu^2 + 2\|B\|^2}{\|B\|^2}\frac{\theta(x^0, z^0, \lambda^0) - \inf\theta_\varphi}{T} + 8\left[2\mu^2 + \frac{\mu^2 + 2\|B\|^2}{\|B\|^2}\chi_1\right]\delta_t,$$
$$\|G_{\alpha_x^{-1}}^{f,\varphi}(x^{t+1}, y^{t+1}, \lambda^{t+1})\|^2 \le 4(1+\alpha_x L_g)^2\left\{\chi_0\frac{\theta(x^0, z^0, \lambda^0) - \inf\theta_\varphi}{T} + \chi_1\delta_t\right\},$$
$$\|G_{\alpha_x^{-1}}^{f,0}(x^{t+1}, y^{t+1}, \lambda^{t+1})\|^2 \le 2\chi_0\frac{\theta(x^0, z^0, \lambda^0) - \inf\theta_\varphi}{T} + 12\chi_1\delta_t$$

and

(4.10)

$$\|G_{\mu}^{f,\psi}(x^{t+1},y^{t+1},\lambda^{t+1})\|^2 \le 8\chi_0 \frac{\mu^2+2\|B\|^2}{\|B\|^2}\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T} + 16\left[2\mu+\frac{\mu^2+2\|B\|^2}{\mu\|B\|^2}\chi_1\right]\Delta_t,$$

$$\|G_{\alpha_x^{-1}}^{f,\varphi}(x^{t+1},y^{t+1},\lambda^{t+1})\|^2 \le 4(1+\alpha_x L_g)^2\left\{\chi_0\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}+\frac{2}{\mu}\chi_1\Delta_t\right\},$$

$$\|G_{\alpha_x^{-1}}^{f,0}(x^{t+1},y^{t+1},\lambda^{t+1})\|^2 \le 12\left[\chi_0\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}+\frac{2}{\mu}\chi_1\Delta_t\right].$$

Proof. We only prove (4.10). The relation (4.9) can be proved similarly. In view of Assumptions 2.1 and 3.1, summing (4.7) over $t=0,1,\dots,T-1$, we obtain

$$\begin{aligned}\sum_{t=0}^{T-1}\|(x^{t+1},z^{t+1},\lambda^{t+1})-(x^t,z^t,\lambda^t)\|^2 &\le \frac{4\alpha_x}{1-\alpha_x L_\theta}[\theta_\varphi(x^0,z^0,\lambda^0)-\theta_\varphi(x^T,z^t,\lambda^T)]+\frac{8\alpha_x^2(\|K\|^2+\mu^2+\|B\|^2)}{\mu(1-\alpha_x L_\theta)^2}\sum_{t=0}^{T-1}\Delta_t\\ &\le \frac{4\alpha_x}{1-\alpha_x L_\theta}[\theta_\varphi(x^0,z^0,\lambda^0)-\inf\theta_\varphi]+\frac{8\alpha_x^2(\|K\|^2+\mu^2+\|B\|^2)}{\mu(1-\alpha_x L_\theta)^2}\sum_{t=0}^{T-1}\Delta_t.\end{aligned}$$

Thus there exists an integer $t\in\{0,1,\dots,T-1\}$ such that

$$\|(x^{t+1},z^{t+1},\lambda^{t+1})-(x^t,z^t,\lambda^t)\|^2 \le \frac{4\alpha_x}{1-\alpha_x L_\theta}\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T} + \frac{8\alpha_x^2(\|K\|^2+\mu^2+\|B\|^2)}{\mu(1-\alpha_x L_\theta)^2}\Delta_t. \tag{4.11}$$

Since

$$y_*(t)=\operatorname{argmax}\left\{f(x^t,y,\lambda^t)-\frac{\mu}{2}\|y-z^t\|^2-\psi(y)\right\},$$

we have

$$0\in-\nabla_y f(x^t,y_*(t),\lambda^t)+\mu[y_*(t)-z^t]+\partial\psi(y_*(t)),$$

implying

$$y_*(t)=\operatorname{prox}_{\mu^{-1}\psi}(z^t+\mu^{-1}\nabla_y f(x^t,y_*(t),\lambda^t)).$$

Thus we get from $z^{t+1}=(1+\alpha_x\mu)z^t-\alpha_x\mu y^{t+1}$ and the nonexpansive property of the proximal mapping that

$$\begin{aligned}&\|G_{\mu}^{f,\psi}(x^{t+1},y^{t+1},\lambda^{t+1})\|\\ &\quad=\|\mu[y^{t+1}-\operatorname{prox}_{\mu^{-1}\psi}(y^{t+1}+\mu^{-1}\nabla_y f(x^{t+1},y^{t+1},\lambda^{t+1}))]\|\\ &\quad=\|\mu[y^{t+1}-y_*(t)]-\mu\operatorname{prox}_{\mu^{-1}\psi}(y^{t+1}+\mu^{-1}\nabla_y f(x^{t+1},y^{t+1},\lambda^{t+1}))\\ &\qquad+\mu\operatorname{prox}_{\mu^{-1}\psi}(z^t+\mu^{-1}\nabla_y f(x^t,y_*(t),\lambda^t))\|\\ &\quad\le\mu\|y^{t+1}-y_*(t)\|+\mu\|y^{t+1}-z^t\|+\|\nabla_y f(x^{t+1},y^{t+1},\lambda^{t+1})-\nabla_y f(x^t,y_*(t),\lambda^t)\|\\ &\quad=\mu\|y^{t+1}-y_*(t)\|+\alpha_x^{-1}\|z^{t+1}-z^t\|\\ &\qquad+\|K^T(x^{t+1}-x^t)+B^T(\lambda^{t+1}-\lambda^t)-[\nabla h(y^{t+1})-\nabla h(y_*(t))]\|\\ &\quad\le(L_h+\mu)\|y^{t+1}-y_*(t)\|+\alpha_x^{-1}\|z^{t+1}-z^t\|+\|B\|\|\lambda^{t+1}-\lambda^t\|+\|K\|\|x^{t+1}-x^t\|,\end{aligned}$$

which implies

$$\begin{aligned}&\|G^{f,\psi}_{\mu^{-1}}(x^{t+1},y^{t+1},\lambda^{t+1})\|^2 \le 4(L_h+\mu)^2\|y^{t+1}-y_*(t)\|^2\\&\quad + 4\max[\|B\|^2,\alpha_x^{-2},\|K\|^2]\|(x^{t+1},z^{t+1},\lambda^{t+1})-(x^t,z^t,\lambda^t)\|^2.\end{aligned}$$

From Theorem 4.2 and the relation (4.11), we obtain

$$\begin{aligned}\|G^{f,\psi}_{\mu^{-1}}(x^{t+1},y^{t+1},\lambda^{t+1})\|^2 \le{}& \frac{8[L_h+\mu]^2}{\mu}\Delta_t\\&+4\max[\|B\|^2,\alpha_x^{-2},\|K\|^2]\left\{\frac{4\alpha_x}{1-\alpha_x L_\theta}\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}\right.\\&\left.+\frac{8\alpha_x^2(\|K\|^2+\mu^2+\|B\|^2)}{\mu(1-\alpha_x L_\theta)^2}\Delta_t\right\}.\end{aligned}$$

From the definition of L_θ and the choice of $\alpha_x\in(0,1/L_\theta)$, we obtain

$$\begin{aligned}&\max[\|B\|^2,\alpha_x^{-2},\|K\|^2]\alpha_x^2 < [\|B\|^2+\alpha_x^{-2}+\|K\|^2]\alpha_x^2 < [\|B\|^2+\|K\|^2]\frac{\mu^2}{\gamma^2}\\&\quad+1\le \frac{\mu^2+2\|B\|^2}{2\|B\|^2}.\end{aligned}$$

Thus we obtain

$$\begin{aligned}&\|G^{f,\psi}_{\alpha_y^{-1}(t)}(x^{t+1},y^{t+1},\lambda^{t+1})\|^2\\&\quad\le \frac{8[L_h+\mu]^2}{\mu}\Delta_t+\frac{2\mu^2+4\|B\|^2}{\|B\|^2}\left\{\frac{4(\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi)}{\alpha_x(1-\alpha_x L_\theta)T}+\frac{8(\|K\|^2+\mu^2+\|B\|^2)}{\mu(1-\alpha_x L_\theta)^2}\Delta_t\right\}\\&\quad\le 16\left[2\mu+\frac{\mu^2+2\|B\|^2}{\mu\|B\|^2}\chi_1\right]\Delta_t+8\chi_0\frac{\mu^2+2\|B\|^2}{\|B\|^2}\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T},\end{aligned}$$

which is just the first inequality in (4.10). From Algorithm 4.1, we can express (x^{t+1},λ^{t+1}) as $(x^{t+1},\lambda^{t+1})=\mathrm{prox}_{\alpha_x\sigma}((x^t,\lambda^t)-\alpha_x\nabla_{x,\lambda}f(x^t,y^{t+1},\lambda^t))$. Thus we have

$$\begin{aligned}&\left\|G^{f,\varphi}_{\alpha_x^{-1}}(x^{t+1},y^{t+1},\lambda^{t+1})\right\|\\&\quad=\left\|\alpha_x^{-1}[x^{t+1}-\mathrm{prox}_{\alpha_x\varphi}(x^{t+1}-\alpha_x\nabla_x f(x^{t+1},\lambda^{t+1},y^{t+1}))\right\|\\&\quad=\alpha_x^{-1}\left\|\mathrm{prox}_{\alpha_x\varphi}(x^t-\alpha_x\nabla_x f(x^t,y^{t+1},\lambda^t))\right.\\&\qquad\left.-\mathrm{prox}_{\alpha_x\varphi}(x^{t+1}-\alpha_x\nabla_x f(x^{t+1},y^{t+1},\lambda^{t+1}))\right\|\\&\quad=\alpha_x^{-1}\left\|\mathrm{prox}_{\alpha_x\varphi}(x^t-\alpha_x[\nabla g(x^t)+Ky^{t+1}+A^T\lambda^t])\right.\\&\qquad\left.-\mathrm{prox}_{\alpha_x\varphi}(x^{t+1}-\alpha_x[\nabla g(x^{t+1})+Ky^{t+1}+A^T\lambda^{t+1}])\right\|\\&\quad\le\alpha_x^{-1}\left\|x^{t+1}-x^t\right\|+L_g\left\|x^{t+1}-x^t\right\|+\|A\|\left\|\lambda^{t+1}-\lambda^t\right\|.\end{aligned}$$

Then we obtain from $\alpha_x^{-2}>2\|A\|^2$, $\max\{(\alpha_x^{-1}+L_g)^2,\|A\|^2\}=(\alpha_x^{-1}+L_g)^2$ that

$$\begin{aligned}&\left\|G^{f,\varphi}_{\alpha_x^{-1}}(x^{t+1},y^{t+1},\lambda^{t+1})\right\|^2\\&\quad\le 2(\alpha_x^{-1}+L_g)^2\|x^{t+1}-x^t\|^2+2\|A\|^2\|\lambda^{t+1}-\lambda^t\|^2\end{aligned}$$

$$
\begin{aligned}
&\leq \max\{(\alpha_x^{-1}+L_g)^2, \|A\|^2\}\left\{\frac{4\alpha_x}{1-\alpha_x L_\theta}\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}\right.\\
&\quad\left.+\frac{8\alpha_x^2(\|K\|^2+\mu^2+\|B\|^2)}{\mu(1-\alpha_x L_\theta)^2}\Delta_t\right\}\\
&\leq (1+\alpha_x L_g)^2\left\{\frac{4}{\alpha_x(1-\alpha_x L_\theta)}\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}+\frac{8(\|K\|^2+\mu^2+\|B\|^2)}{\mu(1-\alpha_x L_\theta)^2}\Delta_t\right\}\\
&= 4(1+\alpha_x L_g)^2\left\{\chi_0\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}+\chi_1\frac{2}{\mu}\Delta_t\right\},
\end{aligned}
$$

which is the second inequality in (4.10). From the updating formula

$$Ax^t + By^{t+1} + c = (\lambda^{t+1}-\lambda^t)/\alpha_x,$$

we have that

$$G^{f,0}_{\alpha_x^{-1}}(x^{t+1}, y^{t+1}, \lambda^{t+1}) = Ax^{t+1} + By^{t+1} + c = A(x^{t+1}-x^t) + \frac{\lambda^{t+1}-\lambda^t}{\alpha_x}.$$

From the relation $\alpha_x^{-2} > 2\|A\|^2$, we obtain from (4.11) that

$$
\begin{aligned}
\|G^{f,0}_{\alpha_x^{-1}}(x^{t+1}, y^{t+1}, \lambda^{t+1})\|^2 &= \|Ax^{t+1} + By^{t+1} + c\|^2\\
&\leq [\|A\|\|x^{t+1}-x^t\| + \alpha_x^{-1}\|\lambda^t-\lambda^{t+1}\|]^2\\
&\leq 2[\|A\|^2\|x^{t+1}-x^t\|^2 + \alpha_x^{-2}\|\lambda^t-\lambda^{t+1}\|^2]\\
&\leq 3\alpha_x^{-2}\|(x^{t+1},\lambda^{t+1})-(x^t,\lambda^t)\|^2\\
&\leq \frac{12}{\alpha_x(1-\alpha_x L_\theta)}\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}\\
&\quad+\frac{24(\|K\|^2+\mu^2+\|B\|^2)}{\mu(1-\alpha_x L_\theta)^2}\Delta_t\\
&= 12\chi_0\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}+\frac{24}{\mu}\chi_1\Delta_t.
\end{aligned}
$$

The proof is completed. ∎

Noting that Proposition 4.4 provides the estimates for $\|G^{f,0}_{\alpha_x^{-1}}(x^{t+1}, y^{t+1}, \lambda^{t+1})\|$, i.e., the estimates for $\|Ax^{t+1}+By^{t+1}+c\|$, we may modify (x^{t+1}, y^{t+1}) to satisfy the equality constraint $Ax + By + c = 0$. This needs the following assumption, which is not hard to satisfy.

Assumption 4.1. Suppose that A and B satisfy that $[A\ B]$ is of full row rank.

It follows from Assumption 4.1 that $AA^T + BB^T$ is positively definite. Assumption 4.1 can be seen as an extension of the linear independent constraint qualification (LICQ) in nonlinear programming, which is a general assumption to ensure existence and uniqueness of Lagrange multipliers, such as [10, 37, 60]. We give the expression of the projection of a point of $\Re^n \times \Re^m$ onto C in the following lemma.

LEMMA 4.5. *Let Assumption* 4.1 *be satisfied. Then for any* $(\widetilde{x}, \widetilde{y})$, *its projection on* C, *denoted by* $\Pi_C(\widetilde{x}, \widetilde{y})$, *is given by*

$$\Pi_C(\widetilde{x}, \widetilde{y}) = (\widetilde{x}, \widetilde{y}) - (A^T\widetilde{\zeta}, B^T\widetilde{\zeta}) \tag{4.12}$$

with $\widetilde{\zeta} = (AA^T + BB^T)^{-1}(A\widetilde{x} + B\widetilde{y} + c)$.

Define

$$(\widetilde{x}^{t+1}, \widetilde{y}^{t+1}) = \Pi_C(x^{t+1}, y^{t+1}). \tag{4.13}$$

Let

$$\omega_y = \|K\|\|A^T[AA^T + BB^T]^{-1}\| + [L_h + 2\mu]\|B^T[AA^T + BB^T]^{-1}\|$$

and

$$\omega_x = [L_g + 2\alpha_x^{-1}]\|A^T[AA^T + BB^T]^{-1}\| + \|K\|\|B^T[AA^T + BB^T]^{-1}\|.$$

Then we have the following conclusion.

PROPOSITION 4.6. *Let Assumptions* 2.1, 3.1, *and* 4.1 *be satisfied. Let* $\alpha_y(t) \in (0, 1/L_h)$, $\alpha_x \in (0, 1/L_\theta)$ *and the sequence* $\{(x^t, z^t, \lambda^t, y^t) : t = 0, 1, \ldots, T\}$ *be generated by Algorithm* 4.1. *Then there exists an integer* $t \in \{0, 1, \ldots, T-1\}$ *such that*

$$\|G^{f,0}_{\alpha_x^{-1}}(\widetilde{x}^{t+1}, \widetilde{y}^{t+1}, \lambda^{t+1})\|^2 = \|A\widetilde{x}^{t+1} + B\widetilde{y}^{t+1} + c\|^2 = 0,$$

$$\begin{aligned}
\|G^{f,\psi}_{\mu}(\widetilde{x}^{t+1}, \widetilde{y}^{t+1}, \lambda^{t+1})\|^2 &\le 8\chi_0\left[\frac{2\mu^2 + 4\|B\|^2}{\|B\|^2} + 3\omega_y^2\right]\frac{\theta(x^0, z^0, \lambda^0) - \inf\theta_\varphi}{T} \\
&\quad + 8\left[4\mu^2 + \frac{2\mu^2 + 4\|B\|^2}{\|B\|^2}\chi_1 + 3\chi_1\omega_y^2\right]\delta_t, \\
\|G^{f,\varphi}_{\alpha_x^{-1}}(\widetilde{x}^{t+1}, \widetilde{y}^{t+1}, \lambda^{t+1})\|^2 &\le 8\chi_0\left[(1 + \alpha_x L_g)^2 + 3\omega_x^2\right]\frac{\theta(x^0, z^0, \lambda^0) - \inf\theta_\varphi}{T} \\
&\quad + 8\chi_1\left[(1 + \alpha_x L_g)^2 + 3\omega_x^2\right]\delta_t
\end{aligned} \tag{4.14}$$

and

$$\begin{aligned}
\|G^{f,\psi}_{\mu}(\widetilde{x}^{t+1}, \widetilde{y}^{t+1}, \lambda^{t+1})\|^2 &\le 8\chi_0\left[\frac{2\mu^2 + 4\|B\|^2}{\|B\|^2} + 3\omega_y^2\right]\frac{\theta(x^0, z^0, \lambda^0) - \inf\theta_\varphi}{T} \\
&\quad + 16\left[4\mu + \frac{2\mu^2 + 4\|B\|^2}{\mu\|B\|^2}\chi_1 + \frac{3}{\mu}\chi_1\omega_y^2\right]\Delta_t, \\
\|G^{f,\varphi}_{\alpha_x^{-1}}(\widetilde{x}^{t+1}, \widetilde{y}^{t+1}, \lambda^{t+1})\|^2 &\le 8\chi_0\left[(1 + \alpha_x L_g)^2 + 3\omega_x^2\right]\frac{\theta(x^0, z^0, \lambda^0) - \inf\theta_\varphi}{T} \\
&\quad + \frac{16}{\mu}\chi_1\left[(1 + \alpha_x L_g)^2 + 3\omega_x^2\right]\Delta_t.
\end{aligned} \tag{4.15}$$

Proof. The equality $\|A\widetilde{x}^{t+1} + B\widetilde{y}^{t+1} + c\|^2 = 0$ comes from the definition (4.13). Here we only prove (4.15), and (4.14) can be proved similarly. Let

$$\widetilde{\zeta}^{t+1} = (AA^T + BB^T)^{-1}(Ax^{t+1} + By^{t+1} + c).$$

Then, from Lemma 4.5, we obtain

$$\begin{aligned}
&\|G^{f,\psi}_{\mu}(\widetilde{x}^{t+1}, \widetilde{y}^{t+1}, \lambda^{t+1})\| \\
&\quad \le \|G^{f,\psi}_{\mu}(x^{t+1}, y^{t+1}, \lambda^{t+1})\| + \|G^{f,\psi}_{\mu}(\widetilde{x}^{t+1}, \widetilde{y}^{t+1}, \lambda^{t+1}) - G^{f,\psi}_{\mu}(x^{t+1}, y^{t+1}, \lambda^{t+1})\| \\
&\quad \le \|G^{f,\psi}_{\mu}(x^{t+1}, y^{t+1}, \lambda^{t+1})\| + 2\mu\|\widetilde{y}^{t+1} - y^{t+1}\| + L_h\|\widetilde{y}^{t+1} - y^{t+1}\| \\
&\qquad + \|K\|\|\widetilde{x}^{t+1} - x^{t+1}\| \\
&\quad = \|G^{f,\psi}_{\mu}(x^{t+1}, y^{t+1}, \lambda^{t+1})\| + (L_h + 2\mu)\|B^T\widetilde{\zeta}^{t+1}\| + \|K\|\|A^T\widetilde{\zeta}^{t+1}\| \\
&\quad \le \|G^{f,\psi}_{\mu}(x^{t+1}, y^{t+1}, \lambda^{t+1})\| + \omega_y\|Ax^{t+1} + By^{t+1} + c\|.
\end{aligned}$$

Thus we have from Proposition 4.4 that

$$
\begin{aligned}
&\|G_\mu^{f,\psi}(\widetilde{x}^{t+1},\widetilde{y}^{t+1},\lambda^{t+1})\|^2\\
&\quad\le 2\|G_\mu^{f,\psi}(x^{t+1},y^{t+1},\lambda^{t+1})\|^2+2\omega_y^2\|Ax^{t+1}+By^{t+1}+c\|^2\\
&\quad\le 32\left[2\mu+\frac{\mu^2+2\|B\|^2}{\mu\|B\|^2}\chi_1\right]\Delta_t+16\chi_0\frac{\mu^2+2\|B\|^2}{\|B\|^2}\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}\\
&\qquad+2\omega_y^2\left\{\frac{12}{\alpha_x(1-\alpha_xL_\theta)}\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}+\frac{24(\|K\|^2+\mu^2+\|B\|^2)}{\mu(1-\alpha_xL_\theta)^2}\Delta_t\right\},
\end{aligned}
$$

which yields the first inequality in (4.15). The second inequality in (4.15) can be proved in the same way. ∎

For developing the iteration complexity of Algorithm 4.1, we need to assume the boundedness of Δ_t. Therefore, we give the following assumption, which ensures the boundedness of y^t.

Assumption 4.2. Suppose that there exists a constant $\beta_1>0$ such that

$$\mathrm{dom}\ \psi\subset\beta_1\mathbf{B},$$

where $\mathrm{dom}\ \psi=\{y:\psi(y)<+\infty\}$ is the effective domain of ψ.

Remark 4.7. If $Y\subset\Re^m$ is a nonempty convex compact set with $Y\subset\beta_1\mathbf{B}$ for some $\beta_1>0$, then

$$\psi(y)=\delta_Y(y)=\begin{cases}0, & y\in Y,\\ +\infty, & y\notin Y,\end{cases}$$

satisfies Assumption 4.2.

Assumption 4.2 is valid in minimax problems over compact convex sets, which have been widely studied in [17, 40, 55, 58].

From the definition of Δ_t in (4.8), assuming the boundedness of $\theta_0(x^t,z^t,\lambda^t)-Q(x^t,z^t,y^t,\lambda^t)$ is another way to ensure the boundedness of Δ_t.

Assumption 4.3. Suppose that there exists a constant $\omega_1>0$ such that

$$\theta_0(x^t,z^t,\lambda^t)-Q(x^t,z^t,y^t,\lambda^t)\le\omega_1$$

for $t=0,1,\dots,T$.

Assumption 4.3 is not easy to verify. However, in some studies on minimax optimization [43], the same assumption is also provided to ensure the complexity of the algorithm.

Define

$$
\begin{aligned}
\gamma_1&=\max\left\{8\left[4\mu^2+\frac{2\mu^2+4\|B\|^2}{\|B\|^2}\chi_1+3\chi_1\omega_y^2\right],8\chi_1\left[(1+\alpha_xL_g)^2+3\omega_x^2\right]\right\},\\
\gamma_2&=\max\left\{8\chi_0\left[\frac{2\mu^2+4\|B\|^2}{\|B\|^2}+3\omega_y^2\right],8\chi_0\left[(1+\alpha_xL_g)^2+3\omega_x^2\right]\right\}.
\end{aligned}
\tag{4.16}
$$

THEOREM 4.8. *Let Assumptions* 2.1, 3.1, *and* 4.1 *be satisfied. Let $N_t\equiv N$ be a constant positive integer for $t\in\mathbf{N}$. Let $\alpha_y(t)\equiv\alpha_y\in(0,1/L_h)$, $\alpha_x\in(0,1/L_\theta)$, and the sequence $\{(x^t,z^t,y^t,\lambda^t):t=0,1,\dots,T\}$ be generated by Algorithm* 4.1. *For any $\epsilon>0$, we have the following conclusions:*

(i) *If Assumption* 4.2 *is satisfied and*

$$N \geq \frac{1}{-\log(1-\mu\alpha_y)}\left[\log(8\gamma_1\beta_1^2)+2\log\frac{1}{\epsilon}\right], \quad T \geq \frac{2\gamma_2(\theta_\varphi(x^0,\lambda^0)-\inf\theta_\varphi)}{\epsilon^2},$$

then there exists an integer $t \in \{0,1,\dots,T-1\}$ *such that* $(\widetilde{x}^{t+1},\widetilde{y}^{t+1},\lambda^{t+1})$ *satisfies*

$$\|G_\mu^{f,\psi}(\widetilde{x}^{t+1},\widetilde{y}^{t+1},\lambda^{t+1})\| \leq \epsilon,\ \|G_{\alpha_x^{-1}}^{f,\varphi}(\widetilde{x}^{t+1},\widetilde{y}^{t+1},\lambda^{t+1})\| \leq \epsilon,\ A\widetilde{x}^{t+1}+B\widetilde{y}^{t+1}+c=0. \tag{4.17}$$

(ii) *If Assumption* 4.3 *is satisfied and*

$$N \geq \frac{1}{-\log(1-\mu\alpha_y)}\left[\log\left(\frac{4\gamma_1\omega_1}{\mu}\right)+2\log\frac{1}{\epsilon}\right],$$
$$T \geq \frac{2\gamma_2(\theta_\varphi(x^0,\lambda^0)-\inf\theta_\varphi)}{\epsilon^2},$$

then there exists an integer $t \in \{0,1,\dots,T-1\}$ *such that* (4.17) *holds.*

Proof. By γ_1 and γ_2 defined by (4.16), $N_t \equiv N \in \mathbf{N}$, $\|y^t - y_*(t)\| \leq 2\beta_1$ by Assumption 4.2, we have from (4.14) of Proposition 4.6 that there exists an integer $t \in \{0,1,\dots, T-1\}$,

$$\|G_{\alpha_x^{-1}}^{f,0}(\widetilde{x}^{t+1},\widetilde{y}^{t+1},\lambda^{t+1})\|^2 = \|A\widetilde{x}^{t+1}+B\widetilde{y}^{t+1}+c\|^2 = 0,$$

$$\|G_\mu^{f,\psi}(\widetilde{x}^{t+1},\widetilde{y}^{t+1},\lambda^{t+1})\|^2 \leq \gamma_2\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}+\gamma_1(1-\mu\alpha_y)^N 4\beta_1^2, \tag{4.18}$$
$$\|G_{\alpha_x^{-1}}^{f,\varphi}(\widetilde{x}^{t+1},\widetilde{y}^{t+1},\lambda^{t+1})\|^2 \leq \gamma_2\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}+\gamma_1(1-\mu\alpha_y)^N 4\beta_1^2.$$

Instead of Assumption 4.2, if Assumption 4.3 holds, then $\theta_0(x^t,\lambda^t)-Q(x^t,y^t,\lambda^t) \leq \omega_1$, and we have from (4.14) of Proposition 4.6 that there exists an integer $t \in \{0,1,\dots, T-1\}$,

$$\|G_\mu^{f,\psi}(\widetilde{x}^{t+1},\widetilde{y}^{t+1},\lambda^{t+1})\|^2 \leq \gamma_2\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}+\frac{2\gamma_1\omega_1}{\mu}(1-\mu\alpha_y)^N; \tag{4.19}$$
$$\|G_{\alpha_x^{-1}}^{f,\varphi}(\widetilde{x}^{t+1},\widetilde{y}^{t+1},\lambda^{t+1})\|^2 \leq \gamma_2\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T}+\frac{2\gamma_1\omega_1}{\mu}(1-\mu\alpha_y)^N.$$

If Assumption 4.2 is satisfied and

$$N \geq \frac{1}{-\log(1-\mu\alpha_y)}\left[\log(8\gamma_1\beta_1^2)+2\log\frac{1}{\epsilon}\right], \quad T \geq \frac{2\gamma_2(\theta_\varphi(x^0,z^0,\lambda^0)-\inf\theta_\varphi)}{\epsilon^2},$$

we have

$$\gamma_2\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T} \leq \frac{\epsilon^2}{2}, \quad (1-\mu\alpha_y)^N 4\gamma_1\beta_1^2 \leq \frac{\epsilon^2}{2}.$$

It follows from this and (4.18) that

$$\|G_{\alpha_y^{-1}}^{f,\psi}(\widetilde{x}^{t+1},\widetilde{y}^{t+1},\lambda^{t+1})\|^2 \leq \varepsilon^2, \quad \|G_{\alpha_x^{-1}}^{f,\varphi}(\widetilde{x}^{t+1},\widetilde{y}^{t+1},\lambda^{t+1})\|^2 \leq \varepsilon^2,$$

which implies the truth of (i). If Assumption 4.3 is satisfied and

$$N \geq \frac{1}{-\log(1-\mu\alpha_y)}\left[\log\left(\frac{4\gamma_1\omega_1}{\mu}\right)+2\log\frac{1}{\epsilon}\right], \quad T \geq \frac{2\gamma_2(\theta_\varphi(x^0,z^0,\lambda^0)-\inf\theta_\varphi)}{\epsilon^2},$$

we have

$$\gamma_2\frac{\theta(x^0,z^0,\lambda^0)-\inf\theta_\varphi}{T} \leq \frac{\epsilon^2}{2}, \quad \frac{2\gamma_1\omega_1}{\mu}(1-\mu\alpha_y)^N \leq \frac{\epsilon^2}{2},$$

which with (4.19) proves (ii). The proof is completed. ∎

Theorem 4.8 tells us, for the structured linearly constrained nonsmooth minimax problem (1.1), that the multistep ascent descent proximal gradient method Algorithm 4.1 can find an ϵ-stationary point in $\mathcal{O}\left(\epsilon^{-2}\log\epsilon^{-1}\right)$ iterations if N and T are chosen as in Theorem 4.8.

The complexity results in Theorem 4.8 of the multistep proximal gradient method Algorithm 4.1 are obtained by reformulating the joint constraint minimax problems into an unconstrained nonsmooth convex-concave minimax problem (2.16). Recently, effective single loop algorithms have been proposed for solving the unconstrained convex-concave nonsmooth minimax problem [13, 29, 52, 64]. Single-step proximal gradient algorithms focus on solving saddle points for the unconstrained minimax problems (1.1), such as those in [13, 29, 31]. Under the assumption of the existence of saddle points, Hamedani [29] proposed a single-step primal-dual algorithm for the unconstrained nonsmooth convex-concave minimax problem. However, in general, $\min\max_{(x,y)\in C} f(x,y) \neq \max\min_{(x,y)\in C} f(x,y)$, where C is defined as in (2.1). Therefore, it is necessary to define the minimax point and stationary points for the joint constraint minimax problems. The definitions of the solution and optimality conditions of the joint constraint smooth minimax problem are established in [23, 34]. To the best of our knowledge, this is the first definition of stationary points (Definition 2.5), and optimality conditions (Lemma 2.4 and Proposition 2.3) are established for the joint constraint nonsmooth minimax problem.

The classical proximal gradient methods are used to solve the nonsmooth minimax problems with $x \in X$, $y \in Y$, where X, Y are convex compact sets, such as those in [5, 51, 52, 64]. A unified single-loop alternating gradient projection algorithm is proposed in [64] for solving smooth nonconvex-concave and convex-nonconcave minimax problems. Assuming the boundedness of Lagrange multipliers, similar to the analysis in Theorem 4.2 in [64], the method can achieve the iteration complexity bound for an ϵ-stationary point in $\mathcal{O}\left(\epsilon^{-2}\right)$ iterations for the nonsmooth convex-concave minimax problem (1.1), where $\mathrm{dom}\varphi$, $\mathrm{dom}\psi$ are bounded sets. For some good problems, it may be possible to determine that the multipliers are bounded. However, if the KKT system of (1.1) is ill-conditioned, namely, the sequence of multipliers increases with iterations when one of the eigenvalues of the gradient of the function in the KKT system is close to 0, the algorithm in [64] is not suitable. In order to overcome the unboundedness of the sequence of multipliers, multistep gradient methods are used for solving the problem, such as in [34]. Therefore, Algorithm 4.1 can be used to solve more general nonsmooth minimax problems.

5. Applications. In this section, we apply the proposed algorithm to solve three classes of problems. The first one is generalized absolute value equations (GAVEs), which can be converted to a smooth convex-concave minimax problem; i.e., $\varphi \equiv 0$ and $\psi \equiv 0$. We discuss the numerical algorithm for solving GAVEs and test the algorithm for three examples. Second, we consider a linear regression (LR) problem, which is a smooth strongly convex strongly concave minimax problem, and we perform the effect of Algorithm 4.1 with different dimensions and constraints. Third, Algorithm 4.1 is used to solve generalized linear projection equations (GLPEs), which are equivalent to a nonsmooth convex-concave minimax problems where φ and ψ are indicator functions. We select projections of three different closed convex cones to report the performance of the algorithm. In this section, all numerical experiments are implemented by MATLAB R2019a on a laptop with Intel(R) Core(TM) i5-6200U 2.30 GHz and 8 GB memory.

We list the algorithms present in the following numerical experiments.

PGmsAD. Proximal gradient multistep ascent descent method in Algorithm 4.1, obtained by iterating multisteps w.r.t. y in the inner subproblem and one-step w.r.t. x and λ in the outer problem.

MGD. Multiplier gradient descent method in [34], obtained by iterating multisteps w.r.t. x and y in the inner subproblem and one-step w.r.t. λ in the outer problem.

PGAD. Proximal gradient ascent descent method in [18] for unconstrained nonsmooth minimax problems.

PIM. Picard iteration method in [57] for GAVE with symmetric positive definite matrix A in (1.2).

TSpcM. Two-step predictor-corrector method in [54] for GAVE with symmetric positive definite matrix A in (1.2).

MN. Modified Newton-type iteration method in [61] for GAVE with positive semi-definite matrix parameter Ω.

SLA. Successive linearization algorithm in [45] for GAVE through solving a linear programming problem with penalty parameter ϵ.

5.1. Generalized absolute value equations. In this part, we focus on generalized absolute value equations (1.2) and prove that GAVE can be translated into a smooth linearly constrainted convex-concave minimax problem. Assume that there exists some $x \in \mathfrak{R}^n$ satisfying GAVE (1.2). So far, some techniques have been used to numerically solve GAVE. One is transforming (1.2) into a concave minimization to obtain a numerical solution, such as in [44, 45, 47]. The second technique for solving (1.2) is using a generalized Newton method [46]. The third one is using the matrix splitting technique to solve (1.2) (see [61, 66]). However, these methods either only obtain the convergence with high probability, or only solve GAVE with special structures, such as $n = m$. For GAVE with $n \neq m$, Alcantara et al. in [35] reformulated GAVE as feasibility equations which they solved via the problem by the method of alternating projections, and proved linear convergence of the algorithm for GAVE with special structures. Different from these numerical algorithms, Algorithm 4.1 can be used to solve GAVE by transforming (1.2) into a convex-concave minimax problem, and find an ϵ-stationary point without any special structure.

In [47], Mangasarian built the equivalence between GAVE and the linear complementarity problem; that is, (1.2) can be rewritten in the following form:

$$A(x^+ - x^-) + B(x^+ + x^-) = b, \quad 0 \le x^+ \perp x^- \ge 0, \tag{5.1}$$

where $x^+ := (\max\{0,\ x_1\}, \ldots, \max\{0,\ x_n\}) \in \mathfrak{R}^n_+$ and $x^- := (\max\{0,\ -x_1\}, \ldots, \max\{0,\ -x_n\}) \in \mathfrak{R}^n_+$. Then, we can obtain the solutions of (5.1) by solving the following linearly constrained minimization:

$$\begin{array}{ll} \min\limits_{x^+,\ x^- \in \Re^n_+} & \langle x^+, x^- \rangle \\ \text{subject to} & A(x^+ - x^-) + B(x^+ + x^-) = b, \end{array}$$

which, by the duality theorem of linear programming, is equivalent to solving the saddle points of the following minimax problem:

$$\begin{array}{ll} \min\limits_{x^+ \in \Re^n_+} \max\limits_{y \in \Re^m} & (b - (A+B)x^+)^T y \\ \text{subject to} & (B-A)^T y \le x^+. \end{array}$$

Obviously, the above problem can be expressed as a linearly constrained minimax problem as follows:

$$\begin{array}{ll} \min\limits_{x^+ \in \Re^n_+} \max\limits_{z \in \Re^n_+, y \in \Re^m} & f(x,y,z) = (b - (A+B)x^+)^T y \\ \text{subject to} & (B-A)^T y + z = x^+. \end{array} \tag{5.2}$$

Then, the proximal gradient multistep ascent descent method can be used for (5.2) as follows.

The solution x^*_{GAVE} to GAVE (1.2) can be expressed through the optimal solution (x^+_*, y_*, z_*) of the convex-concave minimax problem (5.2) as follows:

$$x^*_{GAVE} = x^+_* - P_{\Re^n_+}(x^-_*),$$

where x^-_* satisfies the following linear equations:

$$(B-A)x = b - (A+B)x^+_*.$$

If the matrix $B - A$ is nonsingular, then we have

$$x^*_{GAVE} = x^+_* - P_{\Re^n_+}((B-A)^{-1}(b - (A+B)x^+_*)). \tag{5.3}$$

From the analysis of Theorem 4.8, we establish the iterative complexity of PGmsAD for GAVE in the following remark.

Remark 5.1. Observe that Assumptions 2.1 and 4.1 hold for (5.2). If we assume that (y^t, z^t) generated by PGmsAD is bounded, then Assumptions 4.3 are satisfied, which implies that PGmsAD can find an ϵ-stationary point in $\mathcal{O}\left(\epsilon^{-2}\log\epsilon^{-1}\right)$ iterations.

Since the problem (5.2) is an equivalent problem of GAVE (1.2), we give the relationship between the stationary point of the reformulated minimax problem (5.2) and the optimal solution of the original GAVE problem in the following lemma.

LEMMA 5.2. *Suppose that the matrix $B - A$ is nonsingular. Let $((x^+)^t, y^t, z^t)$ generated by PGmsAD be an ϵ-stationary point of* (5.2) *and $x^t_{GAVE} = (x^+_{GAVE})^t - (x^-_{GAVE})^t$ is defined as*

$$(x^+_{GAVE})^t = \widetilde{(x^+)^t}, \quad (x^-_{GAVE})^t = P_{\Re^n_+}((B-A)^{-1}(b - (A+B)\widetilde{(x^+)^t})). \tag{5.4}$$

Then we have the following conclusions:

(i) *If $x^{t+1}_{GAVE} = x^t_{GAVE}$ for some $t > 0$ holds, then x^t_{GAVE} is the optimal solution of GAVE, i.e., $x^t_{GAVE} = x^*_{GAVE}$.*

(ii) *The sequence x^t_{GAVE} is convergent to the optimal solution of GAVE.*

Proof.

(i) From the strong convexity of $G^{f,\varphi}_{\alpha_x^{-1}}$ defined in (4.17) and the iteration of x in PGmsAD, we have that

$$G^{f,\delta_{\Re^n_+}}_{\alpha_x^{-1}}(\widetilde{(x^+)^t},\widetilde{y}^t,\widetilde{z}^t,\lambda^t)=0 \text{ if and only if } (\widetilde{(x^+)^t},\widetilde{y}^t,\widetilde{z}^t,\lambda^t) = (\widetilde{(x^+)^{t+1}},\widetilde{y}^{t+1},\widetilde{z}^{t+1},\lambda^{t+1}),$$

i.e., $(\widetilde{(x^+)^t},\widetilde{y}^t,\widetilde{z}^t,\lambda^t)=((x^+)^*,y^*,z^*,\lambda^*)$ is a stationary point of (5.2) if and only if $(\widetilde{(x^+)^t},\widetilde{y}^t,\widetilde{z}^t,\lambda^t)=(\widetilde{(x^+)^{t+1}},\widetilde{y}^{t+1},\widetilde{z}^{t+1},\lambda^{t+1})$, where λ^* is the Lagrange multiplier of (5.2). It follows from (5.3) that $(x^+)^*=x^+_*$, which implies that if $\widetilde{(x^+)^t}=\widetilde{(x^+)^{t+1}}$, then

$$\widetilde{(x^+)^t}=(x^+)^*=x^+_*.$$

Hence, from the definition x^*_{GAVE} in (5.3) and x^t_{GAVE} in (5.4), we obtain that if $x^{t+1}_{GAVE}=x^t_{GAVE}$, then $x^t_{GAVE}=x^*_{GAVE}$, which establishes part (i).

(ii) From (5.4), we have that

$$\begin{aligned}
&\|x^{t+1}_{GAVE}-x^t_{GAVE}\| \\
&\quad\le \|\widetilde{(x^+)^{t+1}}-\widetilde{(x^+)^t}\|+\|P_{\Re^n_+}((B-A)^{-1}(b-(A+B)\widetilde{(x^+)^{t+1}})) \\
&\qquad - P_{\Re^n_+}((B-A)^{-1}(b-(A+B)\widetilde{(x^+)^t}))\| \\
&\quad\le \|\widetilde{(x^+)^{t+1}}-\widetilde{(x^+)^t}\|+\|(B-A)^{-1}\|\|(A+B)\|\|\widetilde{(x^+)^{t+1}}-\widetilde{(x^+)^t}\| \\
&\quad\le M_{GAVE}\epsilon,
\end{aligned}$$

where $M_{GAVE}=1+\|(B-A)^{-1}\|\|(A+B)\|$. This implies that $\|x^{t+1}_{GAVE}-x^t_{GAVE}\|\to 0$ as $t\to\infty$, which yields that the sequence x^t_{GAVE} is convergent. Let x^t_{GAVE} converge to $\bar{x}_{GAVE}$. From the definition of x^t_{GAVE} in (5.4), $\bar{x}_{GAVE}$ is the optimal solution of GAVE. ∎

Lemma 5.2 shows that x^t_{GAVE} is convergent, and the limit x^*_{GAVE} satisfies $\|Ax^*_{GAVE}+B|x^*_{GAVE}|-b\|=0$. Combined with Remark 5.1, PGmsAD can find an ϵ-stationary point of (5.2) within $\mathcal{O}\left(\epsilon^{-2}\log\epsilon^{-1}\right)$ iterations.

We tested four examples of GAVE compared with the previous GAVE and minimax algorithms. In the first and second examples, $A,\ B$ are symmetric square matrices and $A+B,\ A-B$ are nonsingular matrices. The third example is a more difficult question where $A,\ B$ are asymmetric square matrices and $A+B,\ A-B$ are singular matrices. Finally, we consider more general linear questions where $A,\ B$ are not square matrices.

(a) $A,B\in\Re^{4\times 4}$ symmetric, A nonsingular, and $A+B, A-B$ nonsingular:

$$A=\begin{pmatrix} 5 & 4 & -3 & 0\\ 4 & 11 & -1 & 0.5\\ -3 & -1 & 6 & 0.5\\ 0 & 0.5 & 0.5 & 0.25\end{pmatrix},$$

$$B=\begin{pmatrix} 2.5 & 1 & -2 & 0\\ 1 & 0.5 & 0.5 & -0.25\\ -2 & 0.5 & 0.5 & -0.25\\ 0 & -0.25 & -0.25 & -0.125\end{pmatrix},\ b=(1,1,1,1)^T.$$

The optimal solution is unique.

(b) $A, B \in \mathfrak{R}^{3\times 3}$ symmetric, A singular, and $A+B, A-B$ nonsingular:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \; B = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \; b = (-1, 4, 1)^T.$$

The optimal solution is $x^* = (1, -1, -1)$ or $(-1, -1, 1)$.

(c) $A, B \in \mathfrak{R}^{3\times 3}$ asymmetric, A nonsingular, and $A+B, A-B$ singular:

$$A = \begin{pmatrix} -0.5 & 0.5 & 1 \\ 0 & 0.5 & 0.5 \\ 0.5 & 1 & 0 \end{pmatrix}, \; B = \begin{pmatrix} -0.5 & 0.5 & 0 \\ -1 & 0.5 & 0.5 \\ 0.5 & 1 & 0 \end{pmatrix}, \; b = (1, 1, 3)^T.$$

The optimal solution is $x^* = (3-2a, a, 4-3a)$ with $0 \le a \le 4/3$.

(d) $A, B \in \mathfrak{R}^{100\times 200}$:

$$A = \begin{pmatrix} 0.01 & \cdots & 0 & \\ & \ddots & & \mathbf{0}_{100\times 100} \\ & 0.01 & 0 & \\ & 0.01 & 0 & \end{pmatrix}^T,$$

$$B = \begin{pmatrix} -1 & -1 & \cdots & -1 & \\ & \ddots & & & -\mathbf{1}_{100\times 100} \\ & & -1 & -1 & \\ & & 0 & -1 & \end{pmatrix}^T,$$

$$b = (-7.99, -7.01, -6, \cdots, -6, -6.01, -5)^T \in \mathfrak{R}^{100}.$$

We denote $\mathbf{0}_{100\times 100}$ and $\mathbf{1}_{100\times 100}$ as matrices with all components of 0 and 1, respectively. Initial points are all chosen as zero vectors.

In Table 1, we report the CPU time and iteration for the best solutions obtained by running seven methods for solving the above four problems. PGmsAD, MGD, and PGAD are valid for GAVE by solving the transformed minimax problem (5.2). Generally, since the minimax algorithms do not depend on the nonsingularity of matrix A, they converge in all test problems. The convergence of GAVE algorithms is faster than that of the minimax algorithms for the problem (a), but PIM, TSpcM, and MN fail in other problems. Even SLA cannot converge in problem (b). Moreover, PGmsAD can find the approximate solution of GAVE within a smaller error than MGD and PGAD in all test problems. In particular, for GAVE (c) with poor matrix structure, PGmsAD achieves lower error within less CPU time compared with MGD and PGAD.

5.2. Linear regression. In this section, we consider the well-known linear regression problem with joint linear constraints as follows:

$$\begin{aligned} &\min_{x\in\Re^n} \max_{y\in\Re^m} && f(x,y) = \tfrac{1}{m}\left[-\tfrac{1}{2}\|y\|^2 - b^T y + y^T K x\right] + \tfrac{\lambda}{2}\|x\|^2 \\ &\text{subject to} && Ax + By + c = 0_p, \end{aligned} \tag{5.5}$$

where the rows of the matrix $K \in \Re^{m\times n}$, $A \in \Re^{p\times n}$, and $B \in \Re^{p\times m}$ are generated by a Gaussian distribution $\mathcal{N}(0, I)$. In the following experiments, let $n = m$, $b = 0$, $c = 0$, and $\lambda = 1/m$.

TABLE 1

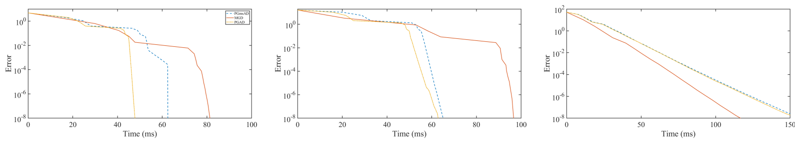
Numerical results of PGmsAD for GAVE (1.2). *(α_x are step sizes w.r.t. x; α_y are step sizes w.r.t. y; α_z are step sizes w.r.t. z; $\alpha_x = \alpha_y = \alpha_z$; N is the number of the inner loop; T is the number of the outer loop; Time is the CPU time; and Error is $\|Ax + B|x| - b\|$).*

Algorithm	Problem	Parameter\Step sizes	N	T	Time (s)	Error
PGmsAD	(a)	$\alpha_x = 0.02$	10	612	8.30e-1	5.77e-2
	(b)	$\alpha_x = 0.05$	5	60	0.24	1.93e-5
	(c)	$\alpha_x = 0.01$	40	390	1.20	5.38e-4
	(d)	$\alpha_x = 0.01$	5	3	9.18e-1	3.32e-12
MGD	(a)	$\alpha_x = 0.00001$	10	5900	2.28	6.10e-2
	(b)	$\alpha_x = 0.001$	50	11	9.4e-2	9.54e-3
	(c)	$\alpha_x = 0.0001$	110	9700	13.64	9.90e-4
	(d)	$\alpha_x = 0.01$	5	3	1.03e-1	3.18e-12
PGAD	(a)	$\alpha_x = 0.02$	\	4789	1.54	2.24e-2
	(b)	$\alpha_x = 0.01$	\	783	3.62e-1	1.32e-3
	(c)	$\alpha_x = 0.0001$	\	7821	1.73	6.28e-4
	(d)	$\alpha_x = 0.01$	\	4	5.01e-2	2.91e-3
PIM	(a)	$\alpha = 0.05$	\	4001	2.02e-1	2.15e-6
	(b)	\	\	\	\	\
	(c)	\	\	\	\	\
	(d)	\	\	\	\	\
TSpcM	(a)	$\alpha = 0.05$	\	4001	3.18e-1	2.24e-6
	(b)	\	\	\	\	\
	(c)	\	\	\	\	\
	(d)	\	\	\	\	\
MN	(a)	$\Omega = I_{4\times 4}$	\	401	5.55e-2	1.13e-10
	(b)	\	\	\	\	\
	(c)	\	\	\	\	\
	(d)	\	\	\	\	\
SLA	(a)	$\epsilon = 0.02$	\	4	8.39	1.49e-15
	(b)	\	\	\	\	\
	(c)	$\epsilon = 0.02$	\	2	5.36	2.08e-15
	(d)	$\epsilon = 0.01$	\	2	6.44	1.92e-9

We compared PGmsAD with MGD and PGAD for solving LR. In both PGmsAD and MGD, the number of inner loops was selected as $N = 3$. We set the step sizes $\alpha_x = 0.3$, $\alpha_y = 1$ and randomly selected the initial points in all algorithms. Algorithms are terminated when at tth iteration, (x^t, y^t, λ^t) is an ϵ-KKT point; i.e., at the tth iteration, $G_L^{f,\varphi}(x^t, y^t, \lambda^t) \le 10^{-7}, G_L^{f,0}(x^t, y^t, \lambda^t) \le 10^{-7}, G_L^{f,\psi}(x^t, y^t, \lambda^t) \le 10^{-7}$.

Figure 1 shows the performance of three algorithms for (5.5) with different dimensions and constraints. As seen from Figure 1, the three algorithms obtain nearly accurate solutions of (5.5). Since LR is a strongly convex strongly concave minimax problem, PGAD can converge for solving (5.5). PGAD shows a faster convergence performance than the other two algorithms in cases (a) and (b), while MGD has better performance in cases (a) and (c). Under the same parameter selection, PGmsAD converges more rapidly than MGD in cases (a) and (b).

5.3. Generalized linear projection equations. In this subsection, we consider the generalized linear projection equations of the form (1.3). One of the classic problems of (1.3) is linear projection equations (e.g., [25, 32, 62]) of the form



(a) $n=10,\ p=10$ (b) $n=100,\ p=40$ (c) $n=1000,\ p=800$

FIG. 1. *The trend of the error for linear regression problems* (5.5) *with respect to CPU time. (Error:* $\|G_L^{f,\varphi}(x^t,y^t,\lambda^t)+G_L^{f,0}(x^t,y^t,\lambda^t)+G_L^{f,\psi}(x^t,y^t,\lambda^t)\|$.*).*

$$P_K(u-Mu-q)=u, \tag{5.6}$$

where $M\in\Re^{n\times n}$, $q\in\Re^n$, and K is a closed polyhedron. Equation (5.6) can be expressed as the projection equations in the following standard form:

$$(I-M)^{-1}x-x_K=-q, \text{ s.t. } x_K=P_K(x).$$

There are some well-established techniques used to solve (5.6), such as Lemke's methods [49], interior-point methods [12], and neural networks [62]. However, there are few numerical algorithms for developing generalized linear projection equations (1.3). In this part, we propose a new technique for solving generalized linear projection equations, where (1.3) is converted to a minimax problem (1.1) and solved with Algorithm 4.1.

In the following, assume that K is a closed convex cone. Hence, for any $x\in\Re^n$, it can be expressed as

$$x=P_K(x)+P_{K^\circ}(x),$$

where K° represents the polar cone of K. Then, (1.3) can be rewritten as

$$\begin{array}{ll}\min\limits_{x_K\in K,x_{K^\circ}\in K^\circ} & \langle x_K,x_{K^\circ}\rangle\\ \text{subject to} & A(x_K+x_{K^\circ})+Bx_K=b,\end{array}$$

where $x_K=P_K(x), x_{K^\circ}=P_{K^\circ}(x)$. By the duality theorem of linear programming, the above problem is equivalent to the following linearly constrained minimax problem:

$$\begin{array}{lll}\min\limits_{x_K\in\Re^n} & \max\limits_{y\in\Re^m,z\in\Re^n} & \delta_K(x_K)+(b-(A+B)x_K)^Ty-\delta_{K^\circ}(z)\\ \text{subject to} & & A^Ty+z=x_K,\end{array} \tag{5.7}$$

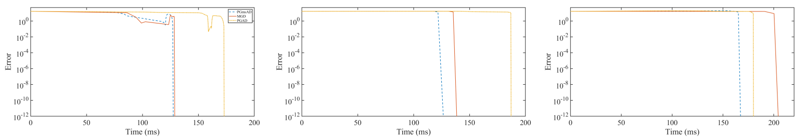
where y is a Lagrange multiplier and z is an auxiliary variable, and $\delta_K(x)$ is an indicator function with $\delta_K(x)=0$ for $x\in K$ while $\delta_K(x)=+\infty$ for $x\notin K$. The solutions to (1.3) can be found by applying PGmsAD to solve (5.7). The solution to GLPE (1.3) can be expressed through the optimal solution (x_K^*,y^*,z^*) of the convex-concave minimax problem (5.7) as follows:

$$x^*_{GLPE}=x^*_K+P_{K^\circ}(x^*_{K^\circ}),$$

where $x^*_{K^\circ}$ satisfies the following linear equations:

$$Ax=b-(A+B)x^*_K.$$

If the matrix A is nonsingular, then we have $x^*_{GLPE}=x^*_K+P_{K^\circ}(A^{-1}(b-(A+B)x^*_K))$. Similar to GAVE, if (y^t,z^t) generated by PGmsAD is bounded and Assumption 3.1



(a) positive octant cone (b) second-order cone (c) 1-norm cone

FIG. 2. *The trend of the error for generalized linear projection equations* (1.3) *under different cones. (Error:* $\|Ax + Bx_K - b\|$*.)*

is satisfied, then results in Theorem 4.8 hold, which implies that PGmsAD can find an ϵ-stationary point in $\mathcal{O}\left(\epsilon^{-2}\log\epsilon^{-1}\right)$ iterations.

We tested projection equations (1.3) with symmetric and asymmetric cones, which are the positive octant cone, second-order cone, and 1-norm cone, respectively expressed as

- positive octant cone (symmetric): $K = \Re^n_+$;
- second-order cone (symmetric): $K = \{x := (s_0; \bar{s}) \in \Re^n : \|\bar{s}\| \le s_0, \bar{s} \in \Re^{n-1}\}$, where $\|\cdot\|$ is the Euclidean norm;
- 1-norm cone (asymmetric): $K = \{x := (s_0; \bar{s}) \in \Re^n : \|\bar{s}\|_1 \le s_0, \bar{s} \in \Re^{n-1}\}$, where $\|\cdot\|_1$ is the 1-norm.

Let nonsingular asymmetric matrices $A, B \in \Re^{n\times n}$ be given as follows:

$$A = \begin{pmatrix} -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 1 & 1 \\ -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 1 & -1 & 1 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0.5 & 0.5 & 1 & 0 & -1 \\ 1 & 0 & 0.5 & 1 & 2 \\ 1 & -1 & 1 & 0.5 & 1 \\ 0 & 0 & -1 & -0.5 & 1 \\ 1 & 0 & 0 & 0 & 0.5 \end{pmatrix},$$

and $b = (6.5, 5, 8.5, -1.5, 8.5)^T$. Let PGmsAD, PMGD, and PGAD be used to solve (5.7) in the above examples, where PMGD combines proximal point methods with MGD, which is a generalization of the multiplier gradient descent method in [34] to nonsmooth minimax problems. In numerical experiments, we set the step sizes $\alpha_x = \alpha_y = \alpha_z = 1/|\det(A+B)|$, and randomly selected the initial points. The number of inner loops was selected as $N = 5$ in PGmsAD and PMGD. Algorithms are terminated when, at the tth iteration, $\|Ax^t + Bx^t_K - b\| \le 10^{-14}$.

The effect of PGmsAD for (1.3) is given in Figure 2. For all symmetric and asymmetric cones, all algorithms obtain nearly accurate solutions of (1.3). PGmsAD achieves lower error within the same CPU time on all cones compared with PMGD and PGAD.

6. Some concluding remarks. Nonsmooth linearly constrained minimax optimization problems are an important class of optimization problems with many applications, such as GAVE, LR, and GLPE. However, there are few numerical algorithms for solving this type of problem if there are joint linear constraints. We developed a conceptual alternating coordinate ascent descent method in which the global convergence is guaranteed if the subproblems with respect to y are solved sufficiently accurately. Combining specific numerical methods (proximal gradient methods) to solve the subproblems, we proposed a proximal gradient multistep ascent descent method and demonstrated the iteration complexity bound for an ϵ-stationary point in $\mathcal{O}\left(\epsilon^{-2}\log\epsilon^{-1}\right)$ iterations under mild conditions. Finally, we applied the proximal gradient multistep ascent descent method to GAVE, LR, and GLPE and proved the linear convergence of the method. In addition, it is worth noting that these numerical algorithms can also be used to solve the linear inequality constrained minimax problem, which can be reformulated as a nonsmooth linear inequality constrained minimax problem by introducing an auxiliary variable z into linear constraints and adding indicator functions $\delta_{\Re^n_+}(z)$ to the objective function.

There are many interesting problems worth considering. Currently, the iteration complexities for the proximal gradient multistep ascent descent method is obtained under Assumption 4.2 or Assumption 4.3. How to guarantee Assumption 4.3 is an important issue for further study. Another issue worth further discussion is how to extend this algorithm to minimax problems with joint nonlinear equality and inequality constraints. Furthermore, another interesting issue is how to use the SA method to solve minimax problems with stochastic joint constraints, where randomness is caused by parameters such as $A,\ B,\ c$.

Appendix A. Preliminaries. The following results are taken from section 10.3 of [6].

LEMMA A.1. *Let $\sigma:\Re^d\to\overline{\Re}$ be a proper lower semicontinuous convex function and $h:\Re^d\to\Re$ be a smooth function whose gradient mapping is Lipschitz continuous with Lipschitz constant $L_h>0$. Then, for $F=\sigma+h$ and $z\in\operatorname{dom}\sigma$, the following hold:*

(a) *The point z^* is a stationary point of F if and only if $G_L^{h,\sigma}(z^*)=0$;*
(b) $\|G_L^{h,\sigma}(z')-G_L^{h,\sigma}(z)\|\le(2L+L_h)\|z'-z\|$;
(c) $F(z)-F(T_L^{h,\sigma}(z))\ge\frac{2L-L_h}{2L^2}\|G_L^{h,\sigma}(z)\|^2$;
(d) $F(z)-F(T_{L_h}^{h,\sigma}(z))\ge\frac{1}{2L_h}\|G_{L_h}^{h,\sigma}(z)\|^2$;
(e) *if in addition h is convex, then for $L\ge L_h$, $z\in\operatorname{dom}\sigma$, $\|G_L^{h,\sigma}(T_L^{h,\sigma}(z))\|\le\|G_L^{h,\sigma}(z)\|$.*

The following lemma is a modified version of Lemma 2 of Bolte [9].

LEMMA A.2. *Let $\sigma:\Re^d\to\overline{\Re}$ be a proper lower semicontinuous function with $\inf_{z\in\Re^d}\sigma(z)>-\infty$, and let $h:\Re^d\to\Re$ be a smooth function whose gradient mapping is Lipschitz continuous with Lipschitz constant $L_h>0$. Then, for $F=\sigma+h$, $z\in\operatorname{dom}\sigma$, $\xi\in\Re^d$, and*

$$z^+\in\operatorname{prox}_{t^{-1}\sigma}\left(z-t^{-1}\xi\right),$$

we have

$$F(z^+)\le F(z)-\frac{1}{2}(t-L_h)\|z^+-z\|^2+\langle\nabla h(z)-\xi,z^+-z\rangle.$$

The following lemma is a modified version of Theorem 10.16 of [6].

LEMMA A.3. *Let $\sigma:\Re^d\to\overline{\Re}$ be a proper lower semicontinuous convex function and $h:\Re^d\to\Re$ be a smooth function whose gradient mapping is Lipschitz continuous with Lipschitz constant $L_h>0$. Then, for $F=\sigma+h$, $z\in\operatorname{dom}\sigma$, $t\ge L_h$, and*

$$z^+=\operatorname{prox}_{t^{-1}\sigma}\left(z-\frac{1}{t}\nabla h(z)\right),$$

we have, for any $z'\in\operatorname{dom}\sigma$,

$$F(z')-F(z^+)\ge\frac{t}{2}\|z'-z^+\|^2-\frac{t}{2}\|z'-z\|^2+l_h(z',z),$$

where

$$l_h(z',z)=h(z')-h(z)-\langle\nabla h(z),z'-z\rangle.$$

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